6.2 Abstract Varieties

In order to deal with affine, quasi-affine, projective and quasi-projective varieties uniformly it is convenient to define an abstract variety. Further some constructions are more natural in this setting (for example product variety). This is analogous to the notion of manifold, and much closer to the notion of scheme. Some differences

a) Sections of the structure sheaf are concrete $\overline{k}$-valued functions rather than abstract elements of a ring in the case of a scheme.
b) When k is not algebraically closed the underlying topological space of the corresponding scheme is the Kolmogorov Quotient (sober-ified to include all the possible generic points). When k is algebraically closed the only difference is the inclusion of generic points.
c) Varieties here are always Noetherian (i.e. there is a finite affine cover and the affines are themselves Noetherian).

The differences are primarily psychological and almost all the theory carries over *mutatis mutandis* to the scheme setting, with additional hypotheses where necessary.

Definition 6.2.1 (Abstract Variety)
*A **variety** over a field k is a space with functions $(X, \mathcal{O}_X)$ such that*

- *There exists a finite open cover $X = \bigcup_{i=1}^n U_i$ together with isomorphisms*
$$\theta_i : \left(U_i, \mathcal{O}_X|_{U_i}\right) \cong \left(X_i\left(\overline{k}\right), \mathcal{O}_{X_i}\right)$$
for some affine varieties $X_1, ..., X_n$ in the sense of (6.1.4).

*The sheaf $\mathcal{O}_X$ is called the **structure sheaf** or **sheaf of regular functions**. A regular morphism is then a morphism of space with functions.*

It is immediate that every affine variety is an abstract variety.

Proposition 6.2.2
A variety $(X, \mathcal{O}_X)$ is a local space with functions. That is for all $f \in \mathcal{O}_X(U)$, the set
$$D(f) := \{x \in U \mid f(x) \neq 0\}$$
is open and $f|_U$ is invertible.

Proof. We have an open affine cover $X = \bigcup_{i=1}^n U_i$. Then we have
$$D(f) = \bigcup_{i=1}^n D\left(f|_{U_i}\right)$$
By (6.1.41) $D\left(f|_{U_i}\right)$ is open and $f|_{U_i}$ is invertible. We deduce that $D(f)$ is open, and using the sheaf conditions that $f|_{D(f)}$ is invertible. ∎

Definition 6.2.3 (Affine (Abstract) Variety)
*Let X be a space of functions. We say it is an **affine variety** if it is isomorphic to $\left(Y\left(\overline{k}\right), \mathcal{O}_Y\right)$, where Y is an affine variety in the sense of (6.1.4) and $\mathcal{O}_Y$ is the corresponding sheaf of regular functions (6.1.25).*

Proposition 6.2.4
Let X be a variety and $f : X \to \overline{k}$ be a function. Then it is regular iff the corresponding map $f : X \to \mathbb{A}^1_k\left(\overline{k}\right)$ is a regular morphism of abstract varieties.

Proof. Suppose that $f : X \to \mathbb{A}^1_k\left(\overline{k}\right)$ is a regular morphism. Evidently the identity map $1 : \mathbb{A}^1_k\left(\overline{k}\right) \to \overline{k}$ is a regular function. Therefore by definition $f : X \to \overline{k}$ is a regular function.

Conversely if $f : X \to \overline{k}$ is regular then so is $f|_U$ for every affine open subset $U \subset X$. By (6.1.29) and (6.1.39) this a regular morphism of abstract varieties $U \to \mathbb{A}^1_k\left(\overline{k}\right)$. By (4.3.27) we have f is a regular morphism. ∎

6.2.1 Topological Properties

Definition 6.2.5 (Integral Variety)
*We say a variety X is **integral** (or irreducible) if it is irreducible as a topological space.*

Proposition 6.2.6 (Criteria for Integrality)
Let X be a variety. Then the following are equivalent

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a) X is integral
b) $\mathcal{O}_X(U)$ is integral for every open subset $U \subset X$

Proposition 6.2.7 (Varieties are Noetherian and Hereditarily Quasi-Compact)
Let X be a variety. Then X is Noetherian as a topological space and every open subset is quasi-compact (and Noetherian).

Proof. (4.1.78) and (4.1.114). ∎

Corollary 6.2.8
Let X be a variety. Then it may be expressed uniquely as a finite union of (closed) irreducible components.

$$X = X_1 \cup ... \cup X_n$$

Proof. See (4.1.76). ∎

Proposition 6.2.9
Suppose $(X, \mathcal{O}_X)$ is a space with functions such that $X = \bigcup_{i=1}^n U_i$ and $\left(U_i, \mathcal{O}_X|_{U_i}\right)$ is a variety. Then so is $(X, \mathcal{O}_X)$.

Proof. Each U_i is a finite union of affine varieties and therefore so is X. ∎

Proposition 6.2.10
Let X be a variety. The (affine) principal open subsets form a base for the topology on X.

Proposition 6.2.11 (Varieties are Symmetric)
Let X be a variety. Then X is symmetric as a topological space and $\overline{\{x\}}$ is finite for all $x \in X$. When k is algebraically closed then every point is closed and therefore X is Kolmogorov.

Furthermore for every open neighbourhood U of x we have $\overline{\{x\}} = \mathrm{cl}_U(\{x\})$.

Proof. That X is symmetric is from (4.1.44) and the affine case (6.1.18). The last statement follows from (4.1.43), then finiteness follows from the affine case. ∎

Proposition 6.2.12 (Closed Subsets of Dimension 0)
Let X be a variety and $Z \subset X$ a closed subset. Then the following are equivalent

a) $\dim Z = 0$
b) $Z = \overline{\{z_1\}} \cup ... \cup \overline{\{z_n\}}$ for some $z_1, ..., z_n \in Z$
c) Z is finite

Proof. Recall X is Noetherian (6.2.7) and therefore by (4.1.91) we have $a) \iff b)$. Then $b) \implies c)$ is (6.2.11) and conversely if Z is finite then $Z = \bigcup_{z \in Z} \overline{\{z\}}$ is a finite union. ∎

Definition 6.2.13 (Separated Variety)
*Let X be a variety. Then we say X is **separated** if for every affine variety Z and pair of morphisms $\phi_1, \phi_2 : Z \rightrightarrows X$ the equaliser*

$$\{z \in Z \mid \phi_1(z) = \phi_2(z)\}$$

is closed in Z.

By (6.1.43) every affine variety is separated.

Proposition 6.2.14
Let X be a separated variety and Z a variety. Then the equaliser of any two maps $\phi_1, \phi_2 : Z \rightrightarrows X$ is closed.

Proof. By definition $Z = \bigcup_{i=1}^n U_i$ where U_i is affine. By assumption the equaliser of $\phi_1|_{U_i}, \phi_2|_{U_i} : U_i \rightrightarrows X$ is closed. As a finite union of closed sets is closed, we are done. ∎

6.2.2 Local Ring

We recall the definition of local ring (4.3.3) for a variety X with irreducible subset W

$$\mathcal{O}_{X,W} := \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U).$$

Note that if $\overline{W'} = \overline{W}$ then $\mathcal{O}_{X,W} = \mathcal{O}_{X,W'}$, so for example $\mathcal{O}_{X,W} = \mathcal{O}_{X,\overline{W}}$.

In the case $W = \{x\}$ we write $\mathcal{O}_{X,x}$ for the local ring and $\mathfrak{m}_{X,x}$ and $k(\mathfrak{m}_{X,x})$ for the maximal ideal and residue field respectively.

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In the case $X = W$ is irreducible then we recover the **field of rational functions** written $k(X)$. For a variety we can make stronger statement about the structure of $k(X)$

Proposition 6.2.15
Let X be an integral variety, $x \in X$ and U an open neighbourhood of x. Then there are canonical injections

$$\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X,x} \hookrightarrow k(X)$$

which makes $k(X)$ the field of fractions for $\mathcal{O}_{X,x}$. When U is affine then $k(X)$ is also the field of fractions for $\mathcal{O}_X(U)$. These maps commute with the restriction maps on $\mathcal{O}_X$, which are also injective.

Proof. By restricting to an affine open cover we may check that the restriction maps ρ_{UV} are injective. This shows that the first map is injective because every stalk can be lifted to some open neighbourhood.

Let U be an affine open neighbourhood of x then by (4.3.16) $k(X) \xrightarrow{\sim} k(U)$. We may deduce from the affine case (6.1.60) that $k(X)$ is the field of fractions for $\mathcal{O}_{X,x}$ and $\mathcal{O}_X(U)$. ∎

As $\mathcal{O}_{X,x}$ is isomorphic to the local ring of an affine neighbourhood we can recover many statements from the affine case.

Proposition 6.2.16
Let X be a variety and $W \subset X$ an irreducible subset. Then the following are equivalent

a) W is contained in only one irreducible component
b) $\mathcal{O}_{X,W}$ is an integral domain

Proof. Firstly we may reduce to the case W is closed by considering $\overline{W}$. As X is a variety, $W \cap U \neq \emptyset$ for U affine. By (4.1.73) we may reduce to the case X is affine, which is (6.1.61). ∎

Proposition 6.2.17 (Dominant Morphisms induce Function Field Maps)
Let X, Y be integral varieties and $\phi : X \to Y$ a dominant regular morphism. Then there is a canonical injective k-algebra homomorphism

$$\begin{aligned} \phi_\star : k(Y) &\hookrightarrow k(X) \\ (V, \sigma) &\to (\phi^{-1}(V), \sigma \circ \phi) \end{aligned}$$

This is functorial in the sense that

$$(\phi \circ \psi)_\star = \psi_\star \circ \phi_\star$$

If ϕ is an open immersion then $\phi_\star$ is an isomorphism.

Proof. See (4.3.7) and (4.3.16). ∎

Proposition 6.2.18 (Function Field Maps are compatible)
Let X, Y be integral varieties, $\phi : X \to Y$ a dominant regular morphism and $y \in \overline{\{\phi(x)\}}$. Then for all neighbourhoods V of y we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(\phi^{-1}(V)) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,\phi(x)} & \xrightarrow{\phi_x} & \mathcal{O}_{X,x} \\ \downarrow & & \downarrow \\ k(Y) & \xrightarrow{\phi_\star} & k(X) \end{array}$$

In particular when identified as subrings of $k(X)$ and $k(Y)$ we have the identities

$$\phi_\star^{-1}(\mathcal{O}_{X,x}) = \mathcal{O}_{Y,\phi(x)}$$

and

$$\phi_\star^{-1}(\mathcal{O}_X(\phi^{-1}(V))) = \mathcal{O}_Y(V)$$

Proof. See (4.3.9) and (6.2.15). ∎

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Lemma 6.2.19
For a variety X, R an algebraic k-algebra and $\phi : \mathcal{O}_{X,x} \to R$ a k-algebra homomorphism Then $\phi^{-1}(\mathfrak{m}) = \mathfrak{m}_{X,x}$ for every maximal ideal $\mathfrak{m} \lhd R$.

Proof. Restatement of (6.1.62), since $\mathcal{O}_{X,x}$ is isomorphic to local ring of affine neighbourhood. ∎

Proposition 6.2.20 (Equivalent Points)
Let X be a variety and $x, y \in X$. Then the following are equivalent

a) x, y are topologically indistinguishable
b) $x \in \overline{\{y\}}$
c) $y \in \overline{\{x\}}$
d) $\overline{\{x\}} = \overline{\{y\}}$
e) $[x] = [y]$ in the Kolmogorov Quotient X_0

When X is integral these are equivalent to $\mathcal{O}_{X,x} = \mathcal{O}_{X,y}$ as subrings of $k(X)$.

Proof. By (6.2.11) X is symmetric and the equivalent conditions a)-e) follow from (4.1.42).

If $x \sim y$ then by definition $\mathcal{O}_{X,x} = \mathcal{O}_{X,y}$.

Conversely suppose $f \in \mathcal{O}_{X,x} \setminus \mathcal{O}_{X,y}$ in $k(X)$. In some open neighbourhood U of x we have $f = \frac{g}{h}$ for $g, h \in \mathcal{O}_X(U)$ and $h(x) \neq 0$. By definition we must have $h(y) = 0$. Therefore $D(h)$ contains x but not y, and therefore x and y are not topologically indistinguishable. ∎

6.2.3 Affine Varieties

We restate various results relating to "abstract" affine varieties to avoid the need to switch notations later on.

Proposition 6.2.21 (Properties of the Structure Sheaf)
Let X be an abstract variety $U \subset X$ an affine open subset. Then the following properties hold

a) For all irreducible subsets $W \subset U$ we have $\mathfrak{M}_{U,W} \coloneqq \{f \in \mathcal{O}_X(U) \mid f|_W = 0\}$ is a prime ideal of $\mathcal{O}_X(U)$. Further we have a canonical local isomorphism

$$\begin{array}{ccc} \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_X(U)_{\mathfrak{M}_{U,W}} \\ & \searrow^{\rho_W} & \sim \downarrow \\ & & \mathcal{O}_{X,W} \end{array}$$

which is the unique homomorphism making this diagram commute.

The inverse image of $\mathfrak{m}_{X,W}$ is $\mathfrak{M}_{U,W}\mathcal{O}_X(U)_{\mathfrak{M}_{U,W}}$ and $\mathfrak{M}_{U,W}$ in $\mathcal{O}_X(U)_{\mathfrak{M}_{U,W}}$ and $\mathcal{O}_X(U)$ respectively.

When $W \subset \overline{\{x\}}$ then $\mathfrak{M}_{U,W} = \mathfrak{M}_{U,x}$ is a maximal ideal.

b) There is a well-defined bijection

$$\begin{aligned} U_0 &\longleftrightarrow \operatorname{Specm}(\mathcal{O}_X(U)) \\ x &\to \mathfrak{M}_{U,x} \end{aligned}$$

Further $\mathfrak{M}_{U,x} = \mathfrak{M}_{U,x'} \iff x \in \overline{\{x'\}} \iff x$ and x' are topologically indistinguishable.

c) For all $f \in \mathcal{O}_X(U)$ we have $\left(D(f), \mathcal{O}_X|_{D(f)}\right)$ is an affine variety. Further we have a canonical isomorphism

$$\begin{array}{ccc} \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_X(U)_f \\ & \searrow^{\rho_{XD(f)}} & \sim \downarrow \\ & & \mathcal{O}_X(D(f)) \end{array}$$

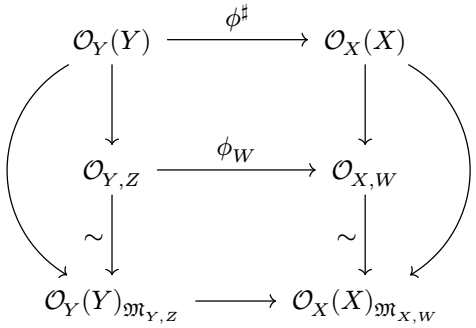
which is the unique homomorphism making this diagram commute. For irreducible $W \subset D(f)$, under this isomorphism the inverse image of $\mathfrak{M}_{D(f),W}$ is $\left(\mathfrak{M}_{U,W}\right)_f$. Similarly the following diagram commutes

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$$\begin{array}{ccc} \mathcal{O}_X(U)_f & \xrightarrow{\sim} & \mathcal{O}_X(D(f)) \\ \downarrow & & \downarrow \\ \mathcal{O}_X(U)_{\mathfrak{M}_{U,W}} & \xrightarrow{\sim} & \mathcal{O}_{X,W} \end{array}$$

and the inverse image of $\mathfrak{m}_{X,W}$ *is* $\mathfrak{M}_{D(f),W}$ *and* $\left(\mathfrak{M}_{U,W}\right)_f$ *in* $\mathcal{O}_X(D(f))$ *and* $\mathcal{O}_X(U)_f$ *respectively.*

d) Suppose X, Y *are affine varieties and* $W \subset X$ *is irreducible and* $Z \subset \overline{\phi(W)}$ *for some irreducible* $Z \subset Y$*. Let* $\phi : X \to Y$ *be a regular morphism. Then the stalk map* $\phi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$ *(see (4.3.5)) makes the following diagram commute*



where the bottom arrow is the localisation map induced by $\phi^\sharp$ *(3.5.22). In fact it is the unique map making either square commute.*

Proposition 6.2.22 (Affine Mapping Property)
Let Y *be an affine variety, and* X *a variety. Then there is a canonical bijection*

$$\begin{aligned} \mathrm{Mor}(X,Y) &\xrightarrow{\sim} \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) \\ \phi &\to \ \phi_Y^\sharp \end{aligned}$$

When X *is affine and* $Z \subset Y$*,* $W \subset X$ *are irreducible subsets, then*

$$Z \subset \overline{\phi(W)} \iff \left(\phi_Y^\sharp\right)^{-1}\left(\mathfrak{M}_{X,W}\right) \subset \mathfrak{M}_{Y,Z}$$

Proof. When X is affine then this is just (6.1.30) and (6.1.32). In general suppose $X = \bigcup_{i=1}^n U_i$ is an affine covering then we have a commutative diagram

$$\begin{array}{ccc} \mathrm{Mor}(X,Y) & \xrightarrow{(-)^\sharp} & \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) \\ \downarrow \alpha & & \downarrow \beta \\ \prod_{i=1}^n \mathrm{Mor}(U_i, Y) & \xrightarrow{\gamma} & \prod_{i=1}^n \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(U_i)) \end{array}$$

where $\alpha(\phi)_i = \phi|_{U_i}$ and $\beta(\psi)_i = \rho_{XU_i} \circ \psi$ and γ is the same definition as $(-)^\sharp$.

By the affine case γ is bijective, and evidently α is injective. Therefore $\beta \circ (-)^\sharp$ is injective, whence $(-)^\sharp$ is injective. To show surjectivity consider $\psi \in \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$ then as γ is bijective we have maps $\phi_i : U_i \to X$ such that $(\phi_i)^\sharp = \rho_{YU_i} \circ \psi$. Suppose $V \subset U_i \cap U_j$ is affine then

$$(\phi_i|_V)^\sharp = \rho_{U_iV} \circ \rho_{XU_i} \circ \psi = \rho_{XV} \circ \psi$$

Therefore $(\phi_i|_V)^\sharp = \left(\phi_j|_V\right)^\sharp$ and from the affine case $\phi_i|_V = \phi_j|_V$. As the open affines from a base (6.2.10) we conclude $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$. Therefore there exists a morphism $\phi : X \to Y$ such that $\phi|_{U_i} = \phi_i$ and therefore $\beta\left(\phi^\sharp\right) = \beta(\psi)$. Evidently β is injective and so $\phi^\sharp = \psi$. ∎

Corollary 6.2.23
There is a full and faithful contravariant functor

$$\mathbf{AbstractAffVar}_k \longrightarrow \mathbf{ReducedFGAlgebras}_k$$

which in particular preserves and reflects isomorphisms

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Corollary 6.2.24
Let X be an affine variety, Y a variety and $f \in \mathcal{O}_X(X)$. Then there is a canonical bijection

$$\begin{aligned} \mathrm{Mor}(D(f), Y) &\xrightarrow{\sim} \mathrm{AlgHom}_k\big(\mathcal{O}_Y(Y), \mathcal{O}_X(X)_f\big) \\ \phi &\to \ \theta_f \circ \phi_Y^\sharp \end{aligned}$$

where $\theta_f : \mathcal{O}_X(D(f)) \xrightarrow{\sim} \mathcal{O}_X(X)_f$ is the isomorphism (6.2.21).c). The image of $i : D(f) \to X$ is simply the canonical localisation map.

Suppose Y is affine and $W \subset D(f)$, $Z \subset Y$ are irreducible subsets. Then $Z \subset \overline{\phi(W)} \iff \big(\theta_f \circ \phi_Y^\sharp\big)^{-1}\Big(\big(\mathfrak{M}_{X,W}\big)_f\Big) \subset \mathfrak{M}_{Y,Z}$.

Proof. We may apply (6.2.22). ∎

Corollary 6.2.25
Let X be an affine variety and $\phi, \psi : X \to Y$ regular morphisms, and $f \in \mathcal{O}_X(X)$. Then $\phi|_{D(f)} = \psi|_{D(f)}$ if and only if $\rho_f \circ \phi^\sharp = \rho_f \circ \psi^\sharp$, where $\rho_f : \mathcal{O}_X(X) \to \mathcal{O}_X(X)_f$ is the canonical localisation map.

Proof. Denote by $i : D(f) \hookrightarrow X$ the inclusion then by (6.2.22) $\phi|_{D(f)} = \psi|_{D(f)} \iff \phi \circ i = \psi \circ i \iff i^\sharp \circ \phi^\sharp = i^\sharp \circ \psi^\sharp$ where $i^\sharp : \mathcal{O}_X(X) \to \mathcal{O}_{D(f)}(D(f))$ is simply the restriction map. We may compose this with the isomorphism (6.2.21).c) to get the required result. ∎

There are some properties of affine varieties which generalise

Proposition 6.2.26
Let X be an affine variety and $f_1, ..., f_r \in \mathcal{O}_X(X)$. Then the following are equivalent

a) $X = \bigcup_{i=1}^r D(f_i)$
b) $(f_1, ..., f_r) = \mathcal{O}_X(X)$

Proof. We may reduce to the case of X a concrete affine variety. Then

$$X = \bigcup_{i=1}^r D(f_i) \iff \emptyset = \bigcap_{i=1}^r V(f_i) = V(f_1, ..., f_r) \iff k[X] = IV(f_1, ..., f_r) = \sqrt{(f_1, ..., f_r)} \iff k[X] = (f_1, ..., f_r)$$

where we have identified $\mathcal{O}_X(X)$ and $k[X]$. ∎

Proposition 6.2.27
Let X be a variety and $f \in \mathcal{O}_X(X)$. There is a canonical isomorphism

$$\mathcal{O}_X(X)_f \xrightarrow{\sim} \mathcal{O}_X(D(f))$$

Proof. The following is adapted from [Liu06] (Prop. 2.3.12).

Recall (6.2.2) that $D(f)$ is an open subset and $f|_{D(f)}$ is invertible. This shows that the homomorphism is well-defined, and we simply need to show it is bijective, with the case X affine already proven (6.1.28).

Let $X = \bigcup_{i=1}^n U_i$ be an affine open covering then $D(f) = \bigcup_{i=1}^n D\big(f|_{U_i}\big)$. By the affine case we have an isomorphism

$$\mathcal{O}_X(U_i)_{f|_{U_i}} \xrightarrow{\sim} \mathcal{O}_X\Big(D\big(f|_{U_i}\big)\Big)$$

By the sheaf conditions we have an exact sequence of $\mathcal{O}_X(X)$-modules

$$0 \longrightarrow \mathcal{O}_X(X) \longrightarrow \bigoplus_{i=1}^n \mathcal{O}_X(U_i) \longrightarrow \bigoplus_{i,j=1}^n \mathcal{O}_X(U_i \cap U_j)$$

Localising at $f \in \mathcal{O}_X(X)$ is exact (3.5.13) and commutes with direct sums (...) so we obtain the exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{O}_X(X)_f & \longrightarrow & \bigoplus_{i=1}^n \mathcal{O}_X(U_i)_{f|_{U_i}} & \longrightarrow & \bigoplus_{i,j=1}^n \mathcal{O}_X(U_i \cap U_j)_{f|_{U_i \cap U_j}} \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & \mathcal{O}_X(D(f)) & \longrightarrow & \bigoplus_{i=1}^n \mathcal{O}_X\Big(D\big(f|_{U_i}\big)\Big) & \longrightarrow & \bigoplus_{i,j=1}^n \mathcal{O}_X\Big(D\big(f|_{U_i \cap U_j}\big)\Big) \end{array}$$

where α, β, γ are just the canonical maps already defined. By the affine case β is an isomorphism. We claim that diagram chasing shows α is injective, for suppose that $\alpha(x) = 0$ then $\alpha(x)|_{D(f|_{U_i})} = 0$ whence $x|_{U_i} = 0$, and by exactness $x = 0$. As $U_i \cap U_j$ is a variety, then we conclude γ is also injective.

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We further claim that α is surjective, for given $y \in \mathcal{O}_X(D(f))$ then there exists $x_i \in \mathcal{O}_X(U_i)_{f_i}$ such that $\beta(x_i) = y|_{D(f|_{U_i})}$. From injectivity of γ and exactness of the bottom sequence we conclude that $x_i|_{U_i \cap U_j} = x_j|_{U_i \cap U_j}$, and so by exactness of the top sequence there is $x \in \mathcal{O}_X(X)_f$ such that $x|_{U_i} = x_i$. Consequently $\alpha(x)|_{D(f|_{U_i})} = y|_{D(f|_{U_i})}$. Exactness of the bottom sequence shows that $\alpha(x) = y$ as required. ∎

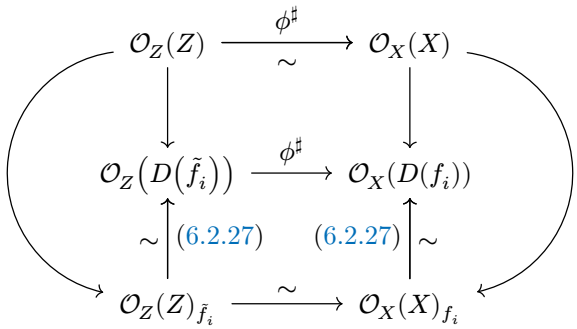
Proposition 6.2.28
Let X be a variety and suppose $f_1, ..., f_r \in \mathcal{O}_X(X)$ generate $\mathcal{O}_X(X)$. If $D(f_i)$ is an affine variety for all $i = 1...r$ then so is X. Furthermore

$$X = \bigcup_{i=1}^{n} D(f_i)$$

Proof. First we claim that $\mathcal{O}_X(X)$ is a reduced finitely generated k-algebra. Recall by (6.2.27) there is a k-algebra isomorphism $\mathcal{O}_X(X)_{f_i} \xrightarrow{\sim} \mathcal{O}_X(D(f_i))$. By assumption $D(f_i)$ is affine so $\mathcal{O}_X(X)_{f_i}$ is finitely-generated as a k-algebra. Therefore by (3.23.21) so is $\mathcal{O}_X(X)$.

By (6.1.36) we may find an affine variety Z together with an isomorphism $\phi^\sharp : \mathcal{O}_Z(Z) \xrightarrow{\sim} \mathcal{O}_X(X)$. By (6.2.22) there is a corresponding morphism of varieties $\phi : X \to Z$, which we wish to show is an isomorphism. Let $\tilde{f}_i := \left(\phi^\sharp\right)^{-1}(f_i)$ then by definition $\phi^{-1}\left(D\left(\tilde{f}_i\right)\right) = D(f_i)$.

We have a commutative diagram



whence the middle arrow is an isomorphism. As $D(f_i)$ is an affine variety we deduce from (6.2.23) that $\phi|_{D(f_i)} : D(f_i) \xrightarrow{\sim} D\left(\tilde{f}_i\right)$ is a regular isomorphism. Further $X = \phi^{-1}(Z)$ whence the $D(f_i)$ cover X and ϕ is a regular isomorphism as required. ∎

6.2.4 Lifting Stalk Maps

In the same way that stalks can be lifted to an open neighbourhood, for varieties we can lift stalk maps to an open neighbourhood. After reducing to the affine case this is essentially an algebraic statement about localisation.

Proposition 6.2.29 (Lifting Stalk Maps I)
Let $\phi, \psi : X \to Y$ be regular morphisms of varieties and $W \subset X$, $Z \subset Y$ non-empty irreducible subsets. Suppose

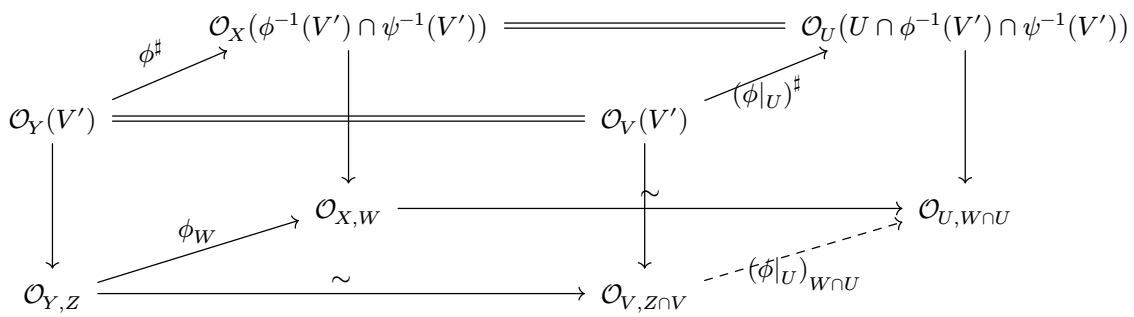
a) $Z \subset \overline{\phi(W)} = \overline{\psi(W)}$, and
b) the stalk maps $\phi_W, \psi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$ agree.

Then there exists an open subset $U \subset X$ for which $U \cap W \neq \emptyset$ and $\phi|_U = \psi|_U$.

Proof. For any open subset $V' \subset Y$ meeting Z (and therefore $\phi(W)$ (4.1.19)), then $\phi^{-1}(V')$ meets W. Similarly for ψ. So by irreducibility $\phi^{-1}(V') \cap \psi^{-1}(V')$ meets W (4.1.60).

Let $V \subset Y$ be an affine open subset meeting Z and choose an affine open subset $U \subset \phi^{-1}(V) \cap \psi^{-1}(V)$ meeting W.

From (4.3.19) we have a commutative cube for every open subset $V' \subset V$



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By uniqueness of the bottom square we conclude that $(\phi|_U)_{W\cap U} = (\psi|_U)_{W\cap U}$ and we may reduce to the case of X and Y affine.

By (6.2.25) it is sufficient to find $f \in \mathcal{O}_X(X)$ such that $\rho_f \circ \phi_Y^\sharp = \rho_f \circ \psi_Y^\sharp$. Suppose that $D(f) \cap W \neq \emptyset$ and consider the commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(Y) & \overset{\phi^\sharp}{\underset{\psi^\sharp}{\longrightarrow}} & \mathcal{O}_X(X) \\ \downarrow & & \downarrow{\scriptstyle \rho_f} \\ & & \mathcal{O}_X(X)_f \\ & & \downarrow \\ \mathcal{O}_Y(Y)_{\mathfrak{m}_{Y,Z}} & \overset{\phi_W}{\underset{\psi_W}{\longrightarrow}} & \mathcal{O}_X(X)_{\mathfrak{m}_{X,W}} \end{array}$$

where we have used (6.2.21) to replace $\mathcal{O}_{Y,Z}$ with $\mathcal{O}_Y(Y)_{\mathfrak{m}_{Y,Z}}$ and $\mathcal{O}_{X,W}$ with $\mathcal{O}_X(X)_{\mathfrak{m}_{X,W}}$.

Let $y_1, ..., y_n$ be generators for $\mathcal{O}_Y$. Then by assumption there exists $t_i \notin \mathfrak{M}_{X,W}$ such that $t_i\phi^\sharp(y_i) = t_i\psi^\sharp(y_i)$. In particular $f = t_1 \cdots t_n$ works, since then $f\phi^\sharp(y_i) = f\psi^\sharp(y_i)$ for all i. ∎

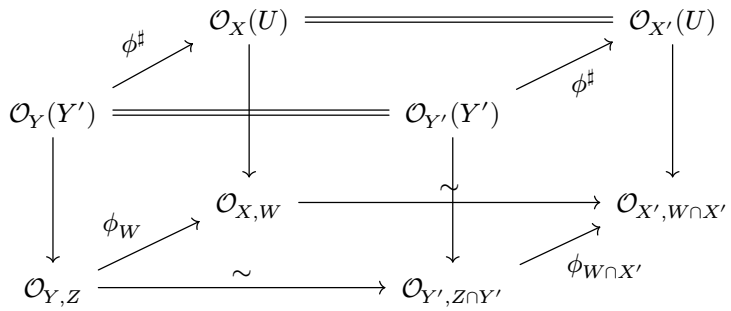
Proposition 6.2.30 (Lifting Stalk Maps II)
Let X, Y be varieties, $W \subset X$, $Z \subset Y$ irreducible subsets and suppose there exists a local k-algebra homomorphism $\underline{\phi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}}$. Then there exists an open subset $U \subset X$ and a regular morphism $\phi : U \to Y$ such that $\overline{Z} = \overline{\phi(W \cap U)}$ and the following diagram commutes for all open subsets $V \subset Y$ meeting Z and $U' \subset \phi^{-1}(V)$ meeting W

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi^\sharp} & \mathcal{O}_X(U') \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,Z} & \xrightarrow[\phi_W]{} & \mathcal{O}_{X,W} \end{array}$$

Proof. Let $Y' \subset Y$ be an open affine subset meeting Z, and $X' \subset \phi^{-1}(V)$ an affine open subset meeting W.

Then $W \cap X'$ is an irreducible subset of X' and similarly for $Z \cap Y'$ (...).

For every open subset $U \subset X'$ meeting W we have a commutative diagram



Therefore we may assume without loss of generality that X and Y are affine, because we may reduce to finding a regular morphism $\phi : U \to Y'$.

We wish to find $f \in \mathcal{O}_X(X)$ and $\phi_Y^\sharp$ which completes the diagram

$$\begin{array}{ccc} \mathcal{O}_Y(Y) & \overset{\phi_Y^\sharp}{\dashrightarrow} & \mathcal{O}_X(X)_f \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,Z} & \xrightarrow{\phi_W^\sharp} & \mathcal{O}_{X,W} \\ \downarrow \sim & & \downarrow \sim \\ \mathcal{O}_Y(Y)_{\mathfrak{m}_{Y,Z}} & \xrightarrow{\phi_W^\sharp} & \mathcal{O}_X(X)_{\mathfrak{m}_{X,W}} \end{array}$$

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For then a regular morphism $\phi : D(f) \to Y$ exists by (6.2.24). and by uniqueness (6.2.21).d) it has the required stalk map. As (6.2.19) $\phi^\sharp_W$ is local we may chase the diagram to show that $\left(\phi^\sharp\right)_Y^{-1}\left(\left(\mathfrak{M}_{X,W}\right)_f\right) = \mathfrak{M}_{Y,Z}$ and therefore $\overline{Z} = \overline{\phi(W \cap D(f))}$ as required.

By inspection it is sufficient to find f such that i_f is injective and $\mathrm{Im}\left(\phi^\sharp_W \circ i_Z\right) \subset \mathrm{Im}\left(i_f\right)$. We may find g such that i_g is injective by (3.5.37). Suppose $\mathcal{O}_Y(Y)$ is generated by $y_1, ..., y_m$ then $\phi^\sharp_W(y_i/1) = x_i/g_i$ for some $g_i \notin \mathfrak{M}_{X,W}$. Then $f = gg_1 \cdots g_m$ satisfies both criteria. As the stalk map is uniquely defined such that the diagram commutes we find that the regular morphism ϕ has stalk map precisely $\phi^\sharp_W$. ∎

Corollary 6.2.31 (Lifting Stalk Maps I)
Let X, Y be varieties, $x \in X$ and $\phi, \psi : X \to Y$ regular morphisms for which

a) $\overline{\{\phi(x)\}} = \overline{\{\psi(x)\}}$
b) $\phi_x = \psi_x$

Then there exists an open neighbourhood U of x such that $\phi|_U = \psi|_U$.

Corollary 6.2.32 (Lifting Stalk Maps II)
Let X, Y be varieties and $\phi^\sharp_x : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ a local k-algebra homomorphism. Then there exists an open subset $U \subset X$ and a regular morphism $\phi : U \to Y$ such that $y \in \overline{\{\phi(x)\}}$ and has stalk map precisely $\phi^\sharp_x$.

6.2.5 Open Subvarieties

Proposition 6.2.33 (Open Subset is Abstract Variety)
*Let X be a variety (resp. separated variety) and $U \subset X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a variety (resp. separated variety), which we call an **open subvariety**. We may write $\mathcal{O}_U \coloneqq \mathcal{O}_X|_U$.*

The inclusion $i : U \hookrightarrow X$ is an open immersion and an open immersion $\phi : X \to Y$ induces an isomorphism between X and the open subvariety $\phi(X)$.

Proof. By definition $X = \bigcup_{i=1}^n U_i$ where U_i is affine. By (...) we have

$$U \cap U_i = \bigcup_{\alpha \in \mathcal{A}_i} D(f_\alpha)$$

where $f_\alpha \in \mathcal{O}_X(U_i)$. By (6.2.21) $\left(D(f_\alpha), \mathcal{O}_X|_{D(f_\alpha)}\right)$ is affine. Therefore we have an open cover of affine sets. By (6.2.7) U is quasi-compact and therefore there exists a finite affine cover. This shows that U is a variety.

Clearly the inclusion map $U \to X$ is injective and regular and so we see that U inherits separatedness. ∎

Proposition 6.2.34 (Quasi-Affine Varieties are Abstract Varieties)
Let X be a quasi-affine variety. Then $(X, \mathcal{O}_X)$ is a separated variety. More precisely there exists a full and faithful functor

$$\mathbf{QuasiAffVar}_k \longrightarrow \mathbf{Var}_k$$

Proof. This follows from (6.2.33) and (6.1.39) ∎

Proposition 6.2.35 (Morphisms into Open Subvarieties)
Let X, Y be varieties and $U \subset Y$ an open subvariety. Then a regular map $\phi : X \to U$ is regular when regarded as a map $\phi : X \to Y$.

Conversely a regular map $\phi : X \to Y$ such that $\phi(X) \subset U$ may be regarded as a regular map $\phi : X \to U$.

Proposition 6.2.36 (Morphisms from Open Subvarieties)
Let $\phi : X \to Y$ be a regular map of varieties and $U \subset X$ an open subvariety. Then $\phi|_U : U \to Y$ is a regular map.

Proposition 6.2.37 (Patching Varieties)
Suppose $X = \bigcup_{i=1}^n U_i$ is a finite union of sets, $\left(U_i, \mathcal{O}_{U_i}\right)$ is a variety, and suppose that for all $i, j = 1...n$

- *$U_i \cap U_j$ is open in U_j, and*
- *the identity map $1 : \left(U_i \cap U_j, \mathcal{O}_{U_i}|_{U_i \cap U_j}\right) \to \left(U_j \cap U_i, \mathcal{O}_{U_j}|_{U_i \cap U_j}\right)$ is a regular morphism of spaces of functions.*

Then there exists a unique structure of a variety on X such that

- *U_i is open in X, and*
- *$\mathcal{O}_{U_i} = \mathcal{O}_X|_{U_i}$.*

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Proposition 6.2.38 (Glueing Varieties)
Suppose $X = \bigcup_{i=1}^{n} U_i$ is a finite union of sets and there exists bijections $\theta_i : U_i \xrightarrow{\sim} W_i$, with W_i varieties such that for all $i, j = 1...n$

- *$\theta_i(U_i \cap U_j)$ is open in W_i, and*
- *$\theta_{ij} = \theta_j \circ \theta_i^{-1} : \theta_i(U_i \cap U_j) \to \theta_j(U_i \cap U_j)$ is regular.*

Then there exists a unique structure of a variety on X such that

- *U_i is open in X, and*
- *θ_i is a regular isomorphism.*

Proof. The topology exists by (4.3.26) and we may simply define

$$\mathcal{O}_X(U) := \left\{ f : U \to \overline{k} \;\middle|\; f \circ \theta_i^{-1}|_{\theta_i(U_i \cap U)} \text{ regular for all } i = 1...n \right\}$$

which gives X the structure of a space of functions. As each U_i is a finite union of affine varieties, so is X. ∎

6.2.6 Closed Subvarieties

Proposition 6.2.39 (Affine Closed Subvarieties)
Let $(X, \mathcal{O}_X)$ be an affine variety with explicit isomorphism $\theta : X \cong V(\mathfrak{a})$, and $Z \subset X$ a closed subset. Then the structure $(Z, \mathcal{O}_Z)$ defined in (4.3.21) is also affine with explicit isomorphism $\theta|_Z : Z \cong V(\mathfrak{b})$ where $\mathfrak{b} := I(\theta(Z))$.

Proof. This follows largely from (6.1.44). ∎

Proposition 6.2.40 (Closed Subvariety)
Let $(X, \mathcal{O}_X)$ be a variety with Z a closed subset. Then $(Z, \mathcal{O}_Z)$ with the structure defined in (4.3.21) is a variety.

Proof. By assumption $X = \bigcup_{i=1}^{n} U_i$ with $\left(U_i, \mathcal{O}_{U_i}\right)$ affine. Evidently $Z = \bigcup_{i=1}^{n} (U_i \cap Z)$ is an open cover. Further $U_i \cap Z$ is a closed subset of U_i. By (6.2.39) $\left(U_i \cap Z, \mathcal{O}_{U_i}|_{U_i \cap Z}\right) \overset{(6.2.47)}{=} \left(U_i \cap Z, \mathcal{O}_Z|_{U_i \cap Z}\right)$ is affine. Therefore Z is a variety. ∎

Definition 6.2.41 (Closed Immersion)
*Let $\phi : X \to Y$ be a regular morphism of varieties. Then it is a **closed immersion** if $\phi(X)$ is closed in Y and ϕ determines an isomorphism between X and the closed subvariety $\phi(X)$.*

Proposition 6.2.42 (Closed Subvariety is a Variety)
Let X be a variety and $Z \subset X$ a closed subset. Then the space with functions $(Z, \mathcal{O}_Z)$ is a variety and the map $j : (Z, \mathcal{O}_Z) \to (X, \mathcal{O}_X)$ is a closed immersion.

When X is affine, then so is Z and the map $j^\sharp : \mathcal{O}_X(X) \to \mathcal{O}_Z(Z)$ is surjective.

Proof. By definition $X = \bigcup_{i=1}^{n} U_i$ where U_i is affine. Then $Z \cap U_i$ is a closed subset of U_i and by (6.2.47) $\mathcal{O}_Z|_{Z \cap U_i} = \mathcal{O}_{U_i}|_{Z \cap U_i}$. Therefore by (6.2.39) $Z \cap U_i$ is affine, and open in Z. This shows that Z is a variety.

Suppose $f \in \mathcal{O}_X(U)$ then $f \circ j \in \mathcal{O}_Z(Z \cap U)$, which shows that j is a regular morphism.

The final statement follows from (6.1.45). ∎

Proposition 6.2.43
Let $\phi : X \to Y$ be a closed immersion and Y an affine variety. Then X is affine.

Proposition 6.2.44
Let $\phi : X \to Y$ be a regular map of varieties and $Z \subset X$ a closed subvariety. Then $\phi|_Z$ is regular.

Proposition 6.2.45
Let $\phi : X \to Y$ be a regular map of varieties and $Z \subset Y$ a closed (resp. open) subvariety. If $\phi(X) \subset Z$ then ϕ may be regarded as a regular map $X \to Z$.

Proposition 6.2.46
Let $\phi : X \to Y$ be a regular map of affine varieties. Then ϕ is a closed immersion if and only if the induced homomorphism $\phi_Y^\sharp : \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$ is surjective.

Proof. Restatement of (6.1.45). ∎

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6.2.7 Locally Closed Subvariety

Proposition 6.2.47 (Locally Closed Subvariety)
Let X be a variety, Z a closed subset and U an open subset. Then $Z \cap U$ is a variety (viewed as a closed subvariety of U or an open subvariety of Z).

6.2.8 Rational Points

For abstract varieties we use a more local definition of rational point which coincides with the affine case (6.1.63).

Definition 6.2.48 (K-Rational Points)
Let X be a variety and K/k an algebraic field extension. Define the K-rational points as follows

$$X(K) \coloneqq \left\{(x, \iota) \;\middle|\; x \in X_0, \quad \iota : k\left(\mathfrak{m}_{X,x}\right) \hookrightarrow K\right\}$$

where X_0 is the Kolmogorov Quotient (4.1.35) of X, and recalling that $\mathcal{O}_{X,x} = \mathcal{O}_{X,y}$ whenever x, y become equal in X_0 (6.2.11).

By (6.2.19) specifying ι is completely equivalent to specifying a k-algebra homomorphism $\mathcal{O}_{X,x} \to K$.

Proposition 6.2.49
Let $\phi : X \to Y$ be a regular morphism of varieties and K/k an algebraic field extension. Then the map

$$\begin{aligned} \phi_K : X(K) &\longrightarrow Y(K) \\ (x, \iota) &\to \quad (\phi(x), \iota \circ \phi_x) \end{aligned}$$

is well-defined and satisfies

$$\psi_K \circ \phi_K = (\psi \circ \phi)_K$$

In particular if ϕ is an isomorphism then ϕ_K is a bijection.

Proposition 6.2.50 (Points = Rational Points over $\overline{k}$)
Let X be a variety. Then there is a canonical bijection

$$\begin{aligned} X &\xrightarrow{\sim} X\left(\overline{k}\right) \\ x &\to \quad ([x], \mathrm{ev}_x) \end{aligned}$$

which is functorial in X. More precisely if $\phi : X \to Y$ is a regular morphism then

$$\mathrm{ev}_{\phi(x)} = \mathrm{ev}_x \circ \overline{\phi}_x$$

Further

$$y = \phi(x) \iff \phi(x) \in \overline{\{y\}} \text{ and } \mathrm{ev}_x \circ \overline{\phi}_x = \mathrm{ev}_y$$

Proof. Recall (4.1.34) that X may be written as a disjoint union

$$X \coloneqq \bigsqcup_{x \in \mathcal{I}} \overline{\{x\}}$$

for some representative subset $\mathcal{I} \subset X$ which is in bijection with X_0. By (6.2.11) X is symmetric and for all $z \in \overline{\{x\}}$ we have $[z] = [x]$ as elements of X_0. So for a given $x \in \mathcal{I}$ we need to show that the map

$$\begin{aligned} \overline{\{x\}} &\to \mathrm{AlgHom}_k\left(k\left(\mathfrak{m}_{X,x}\right), \overline{k}\right) \\ z &\to \mathrm{ev}_z \end{aligned}$$

is bijective. It is well-defined because $z \in \overline{\{x\}}$ implies $\mathcal{O}_{X,x} = \mathcal{O}_{X,z}$. Let U be an affine open neighbourhood of x then by (4.1.43) $\overline{\{x\}} = \mathrm{cl}_U(\{x\})$. Further by (4.3.15) $U \hookrightarrow X$ induces a local isomorphism on stalks and therefore residue fields $k\left(\mathfrak{m}_{X,x}\right) \cong k\left(\mathfrak{m}_{U,x}\right)$. By functoriality (4.3.10) we may then reduce to the case of X affine. The result then follows from the affine case (6.1.63).

There is a commutative diagram

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$$\begin{array}{ccc} X & \xrightarrow{\sim} & X\left(\overline{k}\right) \\ \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle \phi_{\overline{k}}} \\ Y & \xrightarrow{\sim} & Y\left(\overline{k}\right) \end{array}$$

from which the final statement follows easily. ∎

Proposition 6.2.51
Let X be a variety. Then the evaluation map

$$\begin{aligned} \mathrm{ev}_X : X &\longrightarrow \mathrm{AlgHom}_k\left(\mathcal{O}_X(X), \overline{k}\right) \\ x \to \quad & f \to f(x) \end{aligned}$$

is natural in X.

When X is affine it is also a bijection.

Proof. More precisely when $\phi : X \to Y$ is a regular morphism of varieties we require to show that the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\mathrm{ev}} & \mathrm{AlgHom}_k\left(\mathcal{O}_X(X), \overline{k}\right) \\ \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle - \circ \phi^\sharp} \\ Y & \xrightarrow{\mathrm{ev}} & \mathrm{AlgHom}_k\left(\mathcal{O}_Y(Y), \overline{k}\right) \end{array}$$

Suppose $g \in \mathcal{O}_Y(Y)$ and $x \in X$ then this amounts to

$$g(\phi(x)) = \phi^\sharp(g)(x)$$

which holds by definition.

Let $\theta : X \cong V(\mathfrak{a})$ be an isomorphism of varieties. Then $\phi^\sharp$ is an isomorphism of k-algebras, and so $- \circ \phi^\sharp$ is a bijection (since functors preserve isomorphisms). The evaluation map for $V(\mathfrak{a})$ is a bijection (6.1.63) and so we deduce it is also for X. ∎

Corollary 6.2.52
Let X be a variety and $U \subset X$ an affine open subset. Then the evaluation map

$$\mathrm{ev}_U : U \longrightarrow \mathrm{AlgHom}_k\left(\mathcal{O}_X(U), \overline{k}\right)$$

is bijective.

Suppose $\phi : X \to Y$ is a regular morphism, U is an affine neighbourhood of x and V is an affine neighbourhood of y such that $U \subset \phi^{-1}(V)$, then

$$\phi(x) = y \iff \mathrm{ev}_U(x) \circ \rho_{\phi^{-1}(V)U} \circ \phi^\sharp = \mathrm{ev}_V(y)$$

6.2.9 Dimension

Definition 6.2.53 (Dimension of a Variety)
*Let X be a variety. Then we define the **dimension** to be the Krull Dimension of X as a topological space.*

Proposition 6.2.54
Let X be an integral variety. Then it is biequidimensional and

$$\dim X = \mathrm{trdeg}_k(k(X))$$

The dimension equals $\dim U$ for every open subset.

Proof. Let U be an affine open subset then by (6.1.52) $\dim U = \mathrm{trdeg}_k(k(U))$ which equals $\mathrm{trdeg}_k(k(X))$ by (4.3.16). Further by (6.1.51) U is biequidimensional and so by (4.1.73) we conclude that X is biequidimensional with the same dimension as U.

Every open subset can be covered by affines so we are done by (4.1.88). ∎

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Corollary 6.2.55
Let X be a variety. Then it is quasi-biequidimensional. Moreover every closed subset $W \subset X$ satisfies the codimension formula

$$\dim X = \dim W + \operatorname{codim}(W, X)$$

If W is irreducible then we also have the identity

$$\dim \mathcal{O}_{X,W} = \operatorname{codim}(W, X) = \dim X - \dim W$$

Proof. Every irreducible component is an integral closed subvariety and therefore biequidimensional (6.2.54). So by definition X is quasi-biequidimensional. The codimension formula is (4.1.87).

Let U be an affine open subset of X meeting W. Then there is an isomorphism $\mathcal{O}_{X,W} \xrightarrow{\sim} \mathcal{O}_{U,W\cap U}$ (4.3.17). Further by (4.1.73) we have $\operatorname{codim}(W, X) = \operatorname{codim}(W \cap U, U)$. Therefore we may deduce from the affine case (6.1.57) that

$$\dim \mathcal{O}_{X,W} = \dim \mathcal{O}_{U,W\cap U} = \operatorname{codim}(W \cap U, U) = \operatorname{codim}(W, X)$$

as required. ∎

Proposition 6.2.56 (Irreducible Closed Subsets of Dimension 0)
Let X be a variety and $Z \subset X$ an irreducible closed subset. Then the following are equivalent

a) $\dim Z = 0$
b) $Z = \overline{\{z\}}$ *for some* $z \in Z$
c) Z *is finite*

In particular for every $x \in X$ we have $\dim \mathcal{O}_{X,x} = \dim X$.

Proof. Follows from (6.2.12). The dimension formula follows from (6.2.55). ∎

Proposition 6.2.57
Let X be a variety. Then there is a bijective correspondence

$$X_0 \longleftrightarrow \{W \subset Z \mid W \text{ closed and irreducible and } \dim W = 0\}$$

Proof. By (6.2.11) and (4.1.94). ∎

6.2.10 Galois Action

The notion of rational point allows us to define a Galois Action which coincides with concrete notions in both the affine and projective cases.

Definition 6.2.58 (Abstract Galois Action)
Let X be an abstract variety and K/k an algebraic field extension. Then there is a well-defined group action

$$\operatorname{Aut}(K/k) \times X(K) \longrightarrow X(K)$$

given by

$$\sigma, (x, \iota) \longrightarrow (x, \sigma \circ \iota)$$

When $K = \overline{k}$ then this naturally gives a well-defined group action

$$\operatorname{Aut}\left(\overline{k}/k\right) \times X \longrightarrow X$$

in light of the bijection $X \xrightarrow{\sim} X\left(\overline{k}\right)$ (6.2.50).

Proposition 6.2.59 (Topological Properties of Galois Action)
Let X be an abstract variety. For every $x \in X$ and $\sigma \in \operatorname{Aut}\left(\overline{k}/k\right)$ we have x and $\sigma(x)$ are topologically indistinguishable. Furthermore the orbit of x is simply $\overline{\{x\}}$, which has order $\left[k\left(\mathfrak{m}_{X,x}\right) : k\right]$.

In particular σ preserves every open set and is therefore a homeomorphism of X. Finally every σ restricts to a well-defined action on X_0 which is simply the identity.

For every affine open subset $U \subset X$ with explicit isomorphism $U \xrightarrow{\sim} V(\mathfrak{a})$. Then the action of $\operatorname{Aut}\left(\overline{k}/k\right)$ coincides with the usual one under this isomorphism.

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6.2.11 Product Variety

We construct the product in the category of varieties, but note that the topology of the product does *not* coincide with the usual product topology.

We first show that the product, if it exists, is unique in concrete sense.

Lemma 6.2.60 (Uniqueness of Product)
Let X, Y be varieties (resp. affine varieties), then there is at most one topology on $X \times Y$ and local space of functions $\mathcal{O}_{X\times Y}$ for which the projection maps are regular and the following map

$$\begin{aligned} \operatorname{Mor}(Z, X \times Y) &\xrightarrow{\sim} \operatorname{Mor}(Z, X) \times \operatorname{Mor}(Z, Y) \\ \phi &\to \ (\pi_X \circ \phi, \pi_Y \circ \phi) \end{aligned}$$

is a well-defined bijection for all varieties (resp. affine varieties) Z, natural in Z.

Proof. Suppose we have two such structures, denote them by $X \times Y$ and $\widetilde{X \times Y}$, then we have a natural bijection

$$\begin{aligned} \operatorname{Mor}(Z, X \times Y) &\to \operatorname{Mor}\left(Z, \widetilde{X \times Y}\right) \\ \phi &\to \phi \end{aligned}$$

which is the identity on the underlying set function because $\phi = \phi' \iff \pi_X \circ \phi = \pi_X \circ \phi'$ and $\pi_Y \circ \phi = \pi_Y \circ \phi'$. By the dual version of the Yoneda Lemma (2.6.58) the identity map $1 : X \times Y \to \widetilde{X \times Y}$ is a regular isomorphism. Therefore the topology is the same and so are the sheaf of functions. ∎

Proposition 6.2.61
Let X, Y be varieties, then the categorical product $X \times Y$ exists, and coincides with the set-theoretic product. More precisely

a) there exists a unique topology on $X \times Y$ and sheaf of functions $\mathcal{O}_{X\times Y}$ such that the projection maps $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ are regular and there is a natural bijection for all varieties Z

$$\begin{aligned} \operatorname{Mor}(Z, X \times Y) &\xrightarrow{\sim} \operatorname{Mor}(Z, X) \times \operatorname{Mor}(Z, Y) \\ \phi &\to \ (\pi_X \circ \phi, \pi_Y \circ \phi) \end{aligned}$$

b) If X and Y are affine (resp. separated), then so is $X \times Y$, and there is an isomorphism

$$\left(\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y)\right)_{\text{red}} \xrightarrow{\sim} \mathcal{O}_{X\times Y}(X \times Y)$$

c) If $U \subset X$ and $V \subset Y$ are open then so is $U \times V$, and $\mathcal{O}_{U\times V} = \mathcal{O}_{X\times Y}|_{U\times V}$

d) If $T \subset X$ and $S \subset Y$ are closed then $T \times S$ is a closed subset of $X \times Y$

Proof. Suppose X, Y are both affine. Then $X \times Y$ is given the structure of an affine variety in (6.1.96). This means the following map

$$\begin{aligned} \operatorname{Mor}(Z, X \times Y) &\to \operatorname{Mor}(Z, X) \times \operatorname{Mor}(Z, Y) \\ \phi &\to (\pi_X \circ \phi, \pi_Y \circ \phi) \end{aligned}$$

is bijective when Z is affine. Consider the case $Z = \bigcup_{i=1}^n U_i$ is a variety. Injectivity is obvious, for surjectivity suppose $f : Z \to X$ and $g : Z \to Y$ are regular. Then $f|_{U_i}$ and $g|_{U_i}$ are regular, and by the affine case so is $f|_{U_i} \times g|_{U_i}$. By (4.3.27) $f \times g$ is also regular.

Consider the general case $X = \bigcup_{i=1}^n U_i$ and $Y = \bigcup_{j=1}^m V_j$ are varieties with given affine coverings. Define $W_{ij} \coloneqq U_i \times V_j$, then by the affine case W_{ij} may be given the structure of an (affine) variety with regular set-theoretic projections $\pi_X^{ij} : W_{ij} \to U_i$ and $\pi_Y^{ij} : W_{ij} \to V_j$ such that the following map is a bijection

$$\operatorname{Mor}\left(Z, W_{ij}\right) \xrightarrow{\sim} \operatorname{Mor}(Z, U_i) \times \operatorname{Mor}\left(Z, V_j\right)$$

for all varieties Z. Then consider $Z = W_{ij} \cap W_{kl} = \left(\pi_X^{kl}\right)^{-1}(U_i \cap U_k) \cap \left(\pi_Y^{kl}\right)^{-1}\left(V_j \cap V_k\right)$ regarded as an open subvariety of W_{kl}. With this structure the projection maps $\pi_X : Z \to U_i$ and $\pi_Y : Z \to V_j$ may be shown to be regular by

- changing the codomain of $\pi_X^{kl} : W_{kl} \to U_k$ (resp. π_Y) to X, (6.2.35)
- restricting the domain to Z (6.2.36),
- changing the codomain to U_i (resp. V_j) (6.2.35)

Consequently the identity map $W_{ij} \cap W_{kl} \to W_{ij}$ is regular and therefore so is $W_{ij} \cap W_{kl} \to W_{ij} \cap W_{kl}$ (6.2.35) when considered as open subvarieties of W_{kl} and W_{ij} respectively. By symmetry this is a regular isomorphism. Therefore

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the W_{ij} satisfy the criteria of (6.2.37) and there exists a unique structure of a variety on $X \times Y$ for which $\mathcal{O}_{X\times Y}|_{W_{ij}} = \mathcal{O}_{W_{ij}}$. By (...) the set-theoretic projections $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ are regular. As before we need to show the following map is bijective

$$\mathrm{Mor}(Z, X \times Y) \to \mathrm{Mor}(Z, X) \times \mathrm{Mor}(Z, Y)$$
$$\phi \to (\pi_X \circ \phi, \pi_Y \circ \phi)$$

with injectivity obvious. Suppose $f : Z \to X$ and $g : Z \to Y$ are regular. Define $T_{ij} \coloneqq f^{-1}(U_i) \cap g^{-1}(V_j)$. Then by (6.2.36) $f_{ij} \coloneqq f|_{T_{ij}}$ and $g_{ij} \coloneqq g|_{T_{ij}}$ are regular. Therefore by the universal property for W_{ij} the maps $f_{ij} \times g_{ij}$ are regular. Clearly they are compatible so these glue to give a regular map $f \times g : Z \to X \times Y$ (6.2.35) (4.3.27) as required. Further $X \times Y = \bigcup_{i,j} W_{ij}$ so it is indeed a variety. Uniqueness of the structure follows from (6.2.60).

For *b*) we already demonstrated that the affine product is also a product in the category of varieties, so it follows from uniqueness. For separatedness this is an easy exercise.

For *c*) we have $U \times V = \pi_X^{-1}(U) \cap \pi_Y^{-1}(V)$ is open, and the universal property naturally restricts to the subset $U \times V$. The identity then follows from uniqueness (6.2.60).

For *d*) we have $T \times S = \pi_X^{-1}(T) \cap \pi_Y^{-1}(S)$ is closed. ∎

Proposition 6.2.62
Let X be an integral variety and Y a geometrically integral affine variety (see (6.1.84)). Then $X \times Y$ is an integral variety.

Proof. By definition $X = \bigcup_{i=1}^n U_i$ for U_i affine. By (6.2.61) $X \times Y = \bigcup_{i=1}^n U_i \times Y$ and $U_i \times Y$ is affine. By (6.1.98) $U_i \times Y$ is integral. By (4.1.59) $U_i \cap U_j \neq \emptyset$ whence $(U_i \times Y) \cap (U_j \times Y) \neq \emptyset$. Therefore by (4.1.74) we conclude that $X \times Y$ is integral. ∎

6.2.12 Separated Varieties

Proposition 6.2.63
Let $\phi_1, \phi_2 : X \rightrightarrows Y$ be regular morphisms of varieties with Y separated. Then the equaliser

$$\{x \in X \mid \phi_1(x) = \phi_2(x)\}$$

is a closed subset of X.

Proof. Let $X = \bigcup_{i=1}^n U_i$ where U_i are affine. Then the equaliser is the finite union of

$$\left\{x \in U_i \;\middle|\; \phi_1|_{U_i}(x) = \phi_2|_{U_i}(x)\right\}$$

which is by definition closed. ∎

The following shows that the separated property is analogous to the Hausdorff property in the classical case (4.1.98).

Proposition 6.2.64
Let X be a variety. Then X is separated if and only if the diagonal

$$\Delta_X \coloneqq \{(x, x) \mid x \in X\}$$

is a closed subvariety of $X \times X$.

Proof. Suppose that Δ_X is closed. Then the equaliser of $\phi_1, \phi_2 : Z \rightrightarrows X$ is equal to $(\phi_1 \times \phi_2)^{-1}(\Delta_X)$. By (6.2.61) $\phi_1 \times \phi_2$ is regular and in particular continuous. We conclude the equaliser is closed.

Conversely Δ_X is the equaliser of $\pi_1, \pi_2 : X \times X \rightrightarrows X$ and so by (6.2.14) is closed. ∎

Proposition 6.2.65
Let X be a separated variety and $Z \subset X$ a closed (resp. open) subvariety. Then Z is separated.

Proof. Then $\Delta_Z = \Delta_X \cap (Z \times Z)$. By (6.2.61) $Z \times Z$ is a closed (resp. open) subvariety of $X \times X$. Then the result follows by (6.2.64). ∎

Lemma 6.2.66
Let $\phi : X \to Y$ be a regular map of varieties, with Y separated. Then the graph

$$\Gamma_\phi = \{(x, \phi(x)) \mid x \in X\}$$

is a closed subvariety of $X \times Y$. Further there is a canonical isomorphism of varieties

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$$\begin{aligned} X &\xrightarrow{\sim} \Gamma_\phi \\ x &\to (x, \phi(x)) \end{aligned}$$

Proof. Consider the regular map

$$\begin{aligned} (\phi \circ \pi_X) \times \pi_Y : X \times Y &\to Y \times Y \\ (x, y) &\to (\phi(x), y) \end{aligned}$$

Then

$$\Gamma_\phi = ((\phi \circ \pi_X) \times \pi_Y)^{-1}(\Delta_Y)$$

which is closed. The map $X \to \Gamma_\phi$ is evidentally regular. Further $\Gamma_\phi \hookrightarrow X \times Y \to X$ is regular by (6.2.42) and (4.3.2). These are mutual inverses and so the last statement follows immediately. ∎

Proposition 6.2.67 (Intersection of Affines is Affine)
Let X be a separated variety and U, V open subsets. Then there is a regular isomorphism followed by a closed immersion

$$\begin{aligned} U \cap V &\xrightarrow{\sim} (U \times V) \cap \Delta_X \hookrightarrow U \times V \\ x &\to \quad (x, x) \end{aligned}$$

In particular when U, V are affine then so is $U \cap V$.

Proof. By (6.2.61) $U \times V$ is an open subvariety of $X \times X$.

$(U \times V) \cap \Delta_X = ((U \cap V) \times V) \cap \Delta_X$ is the graph of the inclusion map $U \cap V \hookrightarrow V$. By (6.2.66) it is a closed subvariety of $(U \cap V) \times V$ and therefore of $U \times V$. The same result shows that the first map is a regular isomorphism.

When U, V are affine, so is $U \times V$. By (6.2.42) so is $(U \times V) \cap \Delta_X$, and therefore $U \cap V$. ∎

Proposition 6.2.68 (Criteria for Separated Variety)
Let X be a variety with an open cover $\bigcup_{i=1}^n U_i$. Then X is separated if and only if $(U_i \times U_j) \cap \Delta_X$ is a closed subvariety of $U_i \times U_j$ for all $i, j = 1...n$.

Proof. Suppose X is separated then $(U_i \times U_j) \cap \Delta_X$ is a closed subvariety by (6.2.67).

Conversely if $(U_i \times U_j) \cap \Delta_X$ is closed for all i, j then by (4.1.22)

$$\Delta_X = \bigcup_{i,j} (U_i \times U_j) \cap \Delta_X$$

is closed, and therefore X is separated by (6.2.64). ∎

Proposition 6.2.69 (Criteria for Separated Variety II)
Let X be a variety with an affine open cover $\bigcup_{i=1}^n U_i$. Then X is separated if and only if $U_i \cap U_j$ is affine and the k-algebra homomorphism

$$\begin{aligned} \mathcal{O}_X(U_i) \otimes_k \mathcal{O}_X(U_j) &\longrightarrow \mathcal{O}_X(U_i \cap U_j) \\ f \otimes g &\longrightarrow f|_{U_i \cap U_j} \cdot g|_{U_i \cap U_j} \end{aligned}$$

is surjective for all $i, j = 1...n$

Proof. Suppose X is separated, then by (6.2.67) $U_i \cap U_j$ is affine and there is a closed immersion

$$U_i \cap U_j \xrightarrow{\sim} (U_i \times U_j) \cap \Delta_X \hookrightarrow U_i \times U_j$$

By (6.2.46) the corresponding map of global sections

$$\begin{aligned} \mathcal{O}_{U_i \times U_j}(U_i \times U_j) &\to \mathcal{O}_{U_i \cap U_j}(U_i \cap U_j) \quad (*) \\ f &\to u \to f(u, u) \end{aligned}$$

is surjective. Recall from (6.2.61) there is a surjective k-algebra homomorphism

$$\begin{aligned} \mathcal{O}_{U_i}(U_i) \otimes_k \mathcal{O}_{U_i}(U_j) &\to \mathcal{O}_{U_i \times U_j}(U_i \times U_j) \\ f \otimes g &\to u \to f(u) \cdot g(u) \end{aligned}$$

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which composes to give the required map.

Conversely suppose that $U_i \cap U_j$ are affine and the given map is surjective then evidentally the map in $(*)$ is surjective. By (6.2.46) this shows that the map $U_i \cap U_j \hookrightarrow U_i \times U_j$ is a closed immersion and in particular $\big(U_i \times U_j\big) \cap \Delta_X$ is a closed subvariety of $\big(U_i \times U_j\big)$. Then we are done by (6.2.68). ∎

6.2.13 Rational Maps

Definition 6.2.70 (Rational Maps)
*Let X, Y be varieties. A **rational map** $\phi : X \longrightarrow Y$ is an equivalence class of regular morphisms*

$$\phi : U \to Y$$

where U is a dense open subset of X. We say $(U, \phi) \sim (U', \phi')$ if there exists a dense open subset $W \subset U \cap U'$ such that $\phi|_W = \phi'|_W$.

*We say the **domain of definition** of a rational map ϕ is the set of points $x \in X$ for which there exists a neighbourhood U of x and a representative (U, ϕ). This is a dense open subset denoted by $\mathrm{dom}(\phi)$.*

Proposition 6.2.71
Let $\phi : X \longrightarrow Y$ be a rational map with Y separated. Then there exists a representative regular morphism $\phi : \mathrm{dom}(\phi) \to Y$.

Proof. It is enough to show that for any two representatives (U, ϕ) and (U', ϕ') we have $\phi|_{U \cap U'} = \phi'|_{U \cap U'}$. By (6.2.63) the equaliser is closed in $U \cap U'$ and by definition contains a dense open subset W. Therefore it equals $U \cap U'$ as required. ∎

Proposition 6.2.72 (Rational Functions are Rational Maps)
Let X be an integral variety. Then there is a bijection

$$k(X) \xrightarrow{\sim} \{f : X \longrightarrow \mathbb{A}^1_k\}$$

and every such map is either dominant or has finite image.

Proof. By (6.2.4) this is well-defined and bijective. Suppose $U \subset X$ is dense then by (4.1.70) it is irreducible. Therefore a regular map $f : U \to \mathbb{A}^1_k\big(\overline{k}\big)$ has irreducible image (4.1.64) and $\overline{f(X)}$ is irreducible (4.1.65) and closed. Clearly $\dim \overline{f(X)} \leq \dim \mathbb{A}^1_k\big(\overline{k}\big) = 1$. If equality holds then by (4.1.83).f) $\overline{f(X)} = \mathbb{A}^1_k$ and by definition f is dominant.

Otherwise $\dim \overline{f(X)} = 0$ whence by (6.2.56) $\overline{f(X)} = \overline{\{y\}}$ for some $y \in \mathbb{A}^1_k\big(\overline{k}\big)$. ∎

Proposition 6.2.73
Let X be an integral variety. Then as subrings of $k(X)$ (...) we have

$$\mathcal{O}_X(U) = \{f \in k(X) \mid U \subset \mathrm{dom}(f)\}$$

and

$$\mathcal{O}_{X,x} = \{f \in k(X) \mid x \in \mathrm{dom}(f)\}$$

Furthermore

$$\mathcal{O}_X(U) = \bigcap_{x \in U} \mathcal{O}_{X,x}$$

Proof. Recall (...) that $\mathbb{A}^1_k$ being affine is separated. Therefore every rational map $f \in k(X)$ has a regular representative in $\tilde{f} \in \mathcal{O}_X(\mathrm{dom}(f))$ by (6.2.71). So if $U \subset \mathrm{dom}(f)$ then $\tilde{f} \sim \tilde{f}|_U$ and so $f \in \mathcal{O}_X(U)$. Conversely if $\tilde{f} \in \mathcal{O}_X(U)$ then $\big(U, \tilde{f}\big)$ is a rational map whose domain contains U.

The second statement is similar, and then the final statement follows trivially. ∎

Proposition 6.2.74 (Dominant Rational Map)
Let X be an integral variety and $[(U, \phi)] : X \longrightarrow Y$ a rational map. Then the following are equivalent

a) $\phi(U)$ is dense in Y
b) $(U', \phi') \sim (U, \phi) \Longrightarrow \phi'(U')$ is dense in Y

*In this case we say that $[(U, \phi)]$ is a **dominant rational map***

Proof. Clearly $b) \Longrightarrow a)$. Conversely suppose $\emptyset \neq W \subset U$ is open, then it is dense by (4.1.59) so $\mathtt{cl}_U(W) = U$ by (4.1.21). Therefore by (4.1.30)

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$$\phi(U) = \phi(\mathtt{cl}_U(W)) \subset \overline{\phi(W)}$$

which means

$$X = \overline{\phi(U)} \subset \overline{\phi(W)}$$

and therefore $\phi(W)$ is dense. In particular by definition we have $W \subset U \cap U'$ such that $\phi(W) = \phi'(W)$. So we conclude that $\phi'(W)$ is dense and so a-fortiori is $\phi'(U')$. ∎

Proposition 6.2.75
Let X, Y, Z be integral varieties and $\phi : X \longrightarrow Y$, $\psi : Y \longrightarrow Z$ rational maps with ϕ dominant. For every equivalence class (U, ϕ) and (V, ψ) we have $U \cap \phi^{-1}(V) \neq \emptyset$ and the map

$$\psi \circ \phi : U \cap \phi^{-1}(V) \to Z$$

is a regular morphism determining a rational map

$$\psi \circ \phi : X \longrightarrow Z$$

which is dominant provided ψ is.

Further this is independent of the choice of equivalence class.

Proof. By definition $\phi(U)$ is dense which means that $\phi(U) \cap V \neq \emptyset \Longrightarrow U \cap \phi^{-1}(V) \neq \emptyset$, which is therefore also dense in X (4.1.59). ∎

Proposition 6.2.76
Let X, Y be integral varieties. Then there is a bijective map

$$\begin{aligned} \{\phi : X \longrightarrow Y \text{ dominant rational maps}\} &\xrightarrow{\sim} \mathrm{AlgHom}_k(k(Y), k(X)) \\ \phi &\to \ \phi_\star \end{aligned}$$

Further $\phi_\star$ is injective and the existence of ϕ implies that $\dim Y \leq \dim X$. Then following are equivalent

a) $\dim Y = \dim X$
b) $k(X)/\phi_\star(k(Y))$ *is algebraic*
c) $k(X)/\phi_\star(k(Y))$ *is finite*

In this case we denote the ***degree*** *by* $\deg(\phi) := [k(X) : \phi_\star(k(Y))]$.

Proof. The map is well-defined by (6.2.72) and (6.2.75), after we verify that the image is a k-algebra homomorphism.

Further it is injective by (6.2.29) and surjective by (6.2.30).

As $k(Y)$ is a field $\phi_\star$ is injective. We claim that $\mathrm{trdeg}_k(k(Y)) \leq \mathrm{trdeg}_k(k(X))$. For a transcendence base for $k(Y)$ is a-fortiori algebraicly independent as a subset of $k(X)$. Therefore the inequality follows from (3.18.140) and (6.2.54).

a) $\Longrightarrow$ *b)* Suppose $\dim X = \dim Y$ and let $\mathcal{B}$ be a transcendence base for Y, then by (3.18.141) $\phi_\star(\mathcal{B})$ is a transcendence base for X. Therefore $k(X)/\phi_\star(k(\mathcal{B}))$ is algebraic

b) $\Longleftrightarrow$ *c)* A-fortiori $k(X)/\phi_\star(k(Y))$ is finitely generated. Therefore the equivalence follows from (3.18.44).

b) $\Longrightarrow$ *a)* We have a-fortiori $k(X)/\phi_\star(k(\mathcal{B}))$ is algebraic whence by (3.18.140) $\dim Y \geq \dim X$, and so $\dim Y = \dim X$. ∎

6.2.14 Tangent Space

Definition 6.2.77
Let X be a variety and $W \subset X$ an irreducible subset. Define the ***tangent space***

$$T_W X := \mathrm{Der}\big(\mathcal{O}_{X,W}, k(\mathfrak{m}_{X,W})\big)$$

and the ***cotangent space***

$$T^\star_W X := \mathrm{Hom}\big(\mathfrak{m}_{X,W}/\mathfrak{m}^2_{X,W}, k(\mathfrak{m}_{X,W})\big)$$

Definition 6.2.78 (Non-Singular Point)
Let X be a variety and $W \subset X$ an irreducible subset. Then we say that X is ***non-singular*** *at W if*

a) W is contained in a unique irreducible component X_α (equivalently $\mathcal{O}_{X,W}$ is an integral domain), and
b) $\dim T_W X = \dim \mathrm{Der}(k(X_\alpha))$

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We say that X is non-singular if it is integral and non-singular at all points.

Definition 6.2.79 (Regular Point)
Let X be a variety and $W \subset X$ an irreducible subset.

*We say X is **regular** at W if*
a) W is contained in a unique irreducible component X_α, and
b) $\dim \mathcal{O}_{X,W} = \dim T_x^\star X$

We say that X is regular if it is integral and regular at all points.

Proposition 6.2.80 (Regularity / Non-Singularity is a local property)
Let X be a variety, $W \subset X$ an irreducible subset and $U \subset X$ an open subset such that $U \cap W \neq \emptyset$. Then there is a canonical isomorphism

$$T_W X \xrightarrow{\sim} T_{W \cap U} U.$$

X is regular (resp. non-singular) at W if and only if U is regular (resp. non-singular) at $U \cap W$.

Proposition 6.2.81
Let X be an integral variety. Then it is regular (resp. non-singular) if it has a regular (resp. non-singular) open cover.

Proposition 6.2.82
*Let X be a variety and $W \subset X$ an irreducible subset such that $k\big(\mathfrak{m}_{X,W}\big)/k$ is **separably generated** (e.g. k perfect). Then there is a canonical isomorphism*

$$T_W X \xrightarrow{\sim} T_W^\star W$$

*In particular when W is contained in a unique irreducible component we have **regular** $\iff$ **non-singular**.*

6.2.15 Completeness

Completeness is analogous to compactness in the Hausdorff case, and many results remain true when complete is replaced with compact and the standard product topology is used (in place of the topology induced by the product variety structure). We show in the next section that every projective variety is complete, so it is quite a natural condition.

Definition 6.2.83
Let X be a separated variety. Then it is

- ***universally closed** if for every variety Y the projection map*

$$\pi_Y : X \times Y \to Y$$

is closed (i.e. the image of a closed subset is closed).

- ***complete** if for every irreducible closed subvariety $Z \subset X$ and valuation ring $k \subset R \subset k(Z)$ there is a point $z \in Z$ such that $\mathcal{O}_{Z,z} \subset R$*

Proposition 6.2.84 (Image of Complete Variety is Closed)
Let $\phi : X \to Y$ be a morphism of varieties where X is universally closed and Y is separated. Then $\phi(X)$ is a closed subvariety of Y.

Proof. By (6.2.66) the graph Γ_ϕ is a closed subset of $X \times Y$. Then $\phi(X)$ is the projection of Γ_ϕ and therefore closed by definition. ∎

Proposition 6.2.85 (Closed Subvariety of Complete Variety is also Complete)
A closed subvariety of a universally closed (resp. complete) variety is universally closed (resp. complete)

Proof. Suppose $Z \subset X$ is closed then by (6.2.61) $Z \times Y$ is a closed subset of $X \times Y$. Further T closed in $Z \times Y$ implies T closed in $X \times Y$ and the result follows.

The complete case is obvious. ∎

Proposition 6.2.86
A separated variety X is universally closed (resp. complete) iff all its irreducible components are universally closed (resp. complete).

Proof. By (4.1.66) every irreducible component is closed, and therefore complete by (6.2.85).

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Conversely suppose $X = X_1 \cup ... \cup X_n$ is the irreducible decomposition (6.2.8). If $T \subset X \times Y$ is closed then $T \cap (X_i \times Y)$ is closed in $X_i \times Y$ and therefore by assumption $\pi_Y(T \cap (X_i \times Y)) = \pi_Y(T) \cap X_i$ is closed. Therefore $\pi_Y(T) = \bigcup_{i=1}^n \pi_Y(T) \cap X_i$ is also closed.

The complete case is straightforward. ∎

Proposition 6.2.87
Let X be a separated variety. To demonstrate universal closedness it is sufficient to show that

$$\pi_Y : X \times Y \to Y$$

is closed for Y affine (or even $Y := \mathbb{A}^n_k$).

*Further it is sufficient to show that $\pi_Y(T)$ is closed for all **irreducible** closed subsets $T \subset X \times Y$.*

Proof. Suppose $Y = \bigcup_{i=1}^n U_i$ for U_i open affines. Then by (6.2.61) $X \times U_i$ is an open cover of $X \times Y$. Let $T \subset X \times Y$ closed then $T \cap (X \times U_i)$ is closed in $X \times U_i$. By assumption $\pi_Y(T \cap (X \times U_i)) = \pi_Y(T) \cap U_i$ is closed in U_i. Therefore by (4.1.22) $\pi_Y(T)$ is closed in Y as required.

Clearly the closed property is preserved under isomorphism of Y, therefore we may assume that $Y = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ is a closed subset for some n and ideal $\mathfrak{a} \lhd k[X_1, ..., X_n]$.

By (6.2.61) $X \times Y$ is a closed subset of $X \times \mathbb{A}^n_k$. There $T \subset X \times Y$ is also closed in $X \times \mathbb{A}^n_k$ and the result follows.

Finally if T is closed then by (...) $T = T_1 \cup ... \cup T_m$ for T_j irreducible closed subsets of T (and therefore of $X \times Y$). Therefore

$$\pi_Y(T) = \pi_Y\left(\bigcup_{j=1}^m T_j\right) = \bigcup_{j=1}^m \pi_Y(T_j)$$

and it suffices to show that $\pi_Y(T_j)$ is closed. ∎

The following is not strictly needed in application to projective case as it's already known however we include it for completeness of the argument.

Lemma 6.2.88
Let X be an integral variety and $R \subset k(X)$ is a valuation ring with local homomorphism $\Phi : R \to \overline{k}$. Suppose that there exists $x \in X$ such that $\mathcal{O}_{X,x} \subset R$, then x may be chosen such that $\Phi|_{\mathcal{O}_{X,x}} \to \overline{k}$ is evaluation at x.

Proof. By (6.2.19) $\Phi|_{\mathcal{O}_{X,x}}$ is local and by (6.2.50) we may choose $x' \in \overline{\{x\}}$ such that $\Phi|_{\mathcal{O}_{X,x}}$ is precisely evaluation at z'. ∎

The following proof is adapted from [Bum98] (Prop. 7.17).

Proposition 6.2.89 (Complete $\implies$ Universally Closed)
A complete variety is universally closed.

Proof. We may consider the projection $\pi_Y : X \times Y \to Y$ for Y irreducible affine by (6.2.87), and demonstrate that $\pi_Y(T)$ is closed for every irreducible closed subset $T \subset X \times Y$.

Consider the closed subvariety $Z := \mathrm{cl}_X(\pi_X(T))$ of X and the irreducible and affine closed subvariety $Y' := \mathrm{cl}_Y(\pi_Y(T))$ of Y. Then evidentally $T = T \cap (Z \times Y')$ and so

$$\pi_Y(T) = \pi_Y|_{Z \times Y'}\ (T \cap (Z \times Y'))$$

Further by (...) $Z \times Y'$ is a closed subvariety of $X \times Y$ and T is an irreducible closed subset of $Z \times Y'$. Therefore for fixed T we may assume without loss of generality that $\pi_Y(T)$ is dense in Y. The composite maps

$$\pi_Y \circ i : T \hookrightarrow Z \times Y \to Y$$

$$\pi_X \circ i : T \hookrightarrow Z \times Y \to Z$$

are regular (6.2.42) and dominant, and we require to prove the first is surjective. Consider the injective algebra homomorphism (6.2.17)

$$(\pi_Y \circ i)_\star : k(Y) \hookrightarrow k(T)$$

For any $y \in Y$ there is a corresponding evaluation homomorphism $\phi : \mathcal{O}_Y(Y) \to \overline{k}$ (6.2.52). By (3.24.11) there is a valuation ring $(\pi_Y \circ i)_\star(\mathcal{O}_Y(Y)) \subset R \subset k(T)$ and a local homomorphism $\Phi : R \to \overline{k}$ such that $\Phi \circ (\pi_Y \circ i)_\star$ extends ϕ.

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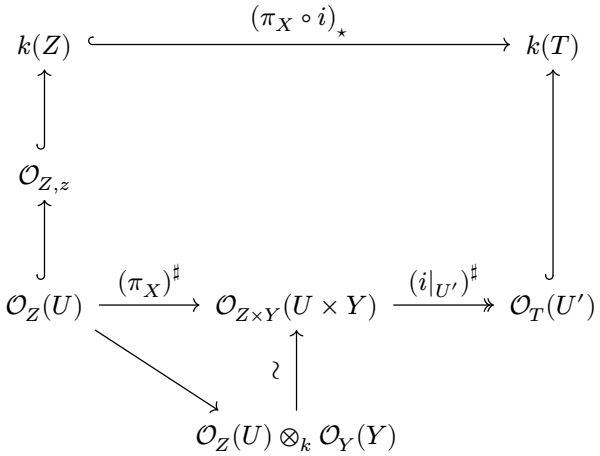
Consider further the injective algebra homomorphism

$$(\pi_X \circ i)_\star : k(Z) \hookrightarrow k(T)$$

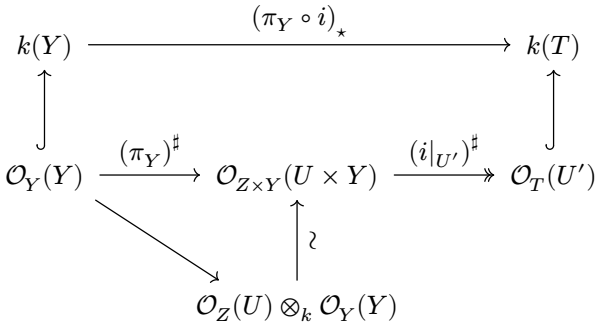
Then $R' \coloneqq (\pi_X \circ i)_\star^{-1}(R)$ is a valuation ring of $k(Z)$ together with local homomorphism $\Phi' : R' \to \overline{k}$ such that $\Phi' = \Phi \circ (\pi_X \circ i)_\star$. By hypothesis there is some $z \in Z$ such that $\mathcal{O}_{Z,z} \subset R'$ and by (6.2.88) we may assume that $\Phi'|_{\mathcal{O}_{Z,z}}$ is precisely evaluation at z.

Let U be an open affine neighbourhood of z in Z, then by (6.2.61) $U \times Y$ is an affine open subset of $Z \times Y$. As $\pi_X(T)$ is dense in Z then $U \cap \pi_X(T)$ is non-empty. This shows that $U' \coloneqq (U \times Y) \cap T$ is a non-empty open subset of T. Evidently U' is a closed subset of $U \times Y$. Therefore it is also affine (6.2.47) (6.2.42).

There is a commutative diagram (6.2.18)



By construction of R we see that $(i|_{U'})^\sharp(\mathcal{O}_Z(U) \otimes 1) \subset R$ (after suitable identifications). Similarly we have a commutative diagram



and by construction we have $(i|_{U'})^\sharp(1 \otimes \mathcal{O}_Y(Y)) \subset R$ (after suitable idenitifications). By properties of the tensor product this means

$$(i|_{U'})^\sharp(\mathcal{O}_Z(U) \otimes \mathcal{O}_Y(Y)) \subset R$$

By (6.2.42) the map $(i|_{U'})^\sharp$ is surjective, therefore we conclude that $\mathcal{O}_T(U') \subset R$. Then by (6.2.52) the restriction of Φ to $\mathcal{O}_T(U')$ corresponds to a point (z', y') of U'. Further $\Phi \circ (\pi_Y \circ i|_{U'})^\sharp = \big(\Phi \circ (\pi_Y \circ i)_\star\big)|_{\mathcal{O}_Y(Y)} = \phi$ and so by (6.2.52) we have $(\pi_Y \circ i)(z', y') = y$ that is to say $y' = y$. This shows that $\pi_Y \circ i$ is surjective as required. ∎

Proposition 6.2.90

Let X, Y be varieties with Y complete such that

a) $U \subset X$ is an irreducible open neighbourhood
b) $x \in \overline{U}$
c) $\mathcal{O}_{X,x}$ is a valuation ring
d) $\phi : U \to Y$ is a regular morphism

Then there exists an open subset $U' \supset U$ for which $\overline{\{x\}} \subset U'$ and ϕ lifts to U'.

Proof. We claim it's enough to find an open neighbourhood V of x such that $U \cap V \neq \emptyset$ and a regular function $\psi : V \to Y$ such that $\psi|_{U\cap V} = \phi|_{U\cap V}$. For then this extends to a regular function $U \cup V$ as required.

Replacing $X = \overline{U}$ we may assume that U is dense in X and X is irreducible. Observe $Z \coloneqq \mathrm{cl}_Y(\phi(U))$ is an irreducible closed subset of Y (4.1.70) (4.1.65). Therefore there is an injective k-algebra homomorphism (4.3.16)

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$$\phi_\star : k(Z) \hookrightarrow k(U) \xrightarrow{\sim} k(X)$$

By assumption $\mathcal{O}_{X,x}$ is a valuation ring, whence so is the inverse image R. By assumption there is $z \in Z$ such that $\mathcal{O}_{Z,z} \subset R$. Therefore we have a k-algebra homomorphism $\mathcal{O}_{Z,z} \to \mathcal{O}_{X,x}$. By (6.2.32) this lifts to a regular morphism $\psi : V \to Z \subset Y$ for which $x \in V$. As $x \in \overline{U}$ we have $U \cap V \neq \emptyset$ by (4.1.19). As Y is separated (6.2.63) the equaliser of ϕ, ψ is a closed subset of $U \cap V$. We have by construction a commutative diagram

$$\begin{array}{ccc} k(Z) & \xrightarrow{\phi_\star} & k(X) \\ \uparrow & & \uparrow \\ \mathcal{O}_{Z,z} & \xrightarrow{\psi_x^\sharp} & \mathcal{O}_{X,x} \\ \uparrow & & \uparrow \\ \mathcal{O}_Z(Z) & \xrightarrow{\psi_Z^\sharp} & \mathcal{O}_X(V) \end{array}$$

Regarding $\psi : X \longrightarrow Y$ as a rational map we find by (6.2.76) that $\psi_\star = \phi_\star$ whence $\psi = \phi$ as rational maps. This means that the equaliser contains a dense open subset of $U \cap V$ and being closed is therefore the whole set. That is $\phi|_{U\cap V} = \psi|_{U\cap V}$ as required. ∎

6.2.16 Projective Space

We introduce the notion of projective varieties by showing that projective space can naturally be given the structure of an abstract variety. A more low-tech approach is taken in the Projective Varieties section, where the constructions are shown to be equivalent.

Definition 6.2.91 (Projective Space)
Let K be a field extension define

$$\mathbb{P}^n(K) := K^{n+1} \setminus \{(0,...,0)\}/\sim$$

where

$$(x_0,...,x_n) \sim (x_{0'},...,x_{n'}) \iff x_i = \lambda x_{i'} \text{ some } \lambda \in K^\star$$

and write $(x_0 : ... : x_n)$ for the corresponding equivalence class in $\mathbb{P}^n(K)$.

Lemma 6.2.92 (Comparing projective points)
Consider two projective points $x = (x_0 : ... : x_n)$ and $y = (y_0 : ... : y_n)$.

a) $x = y \implies \forall i (x_i = 0 \iff y_i = 0)$

Suppose that $x_i \neq 0$ for some i. Then $x = y$ iff

a) $y_i \neq 0$
b) $\frac{x_j}{x_i} = \frac{y_j}{y_i}$ for all j

Proposition 6.2.93 (Projective Space)
Consider the set $X = \mathbb{P}^n\left(\overline{k}\right)$ and for every $i = 0,...,n$ define the subset

$$U_i := \left\{(x_0 : ... : x_n) \;\middle|\; x_0,...,x_n \in \overline{k}, x_i \neq 0\right\}.$$

There are canonical bijections

$$\begin{aligned} \theta_i : U_i &\xrightarrow{\sim} \mathbb{A}_k^n\left(\overline{k}\right) \quad i = 0...n \\ (x_0 : ... : x_n) &\to \left(\frac{x_0}{x_i},...,\frac{\widehat{x_i}}{x_i},...,\frac{x_n}{x_i}\right) \\ (z_1 : ... : 1 : ... : z_n) &\leftarrow (z_1,...,z_n) \end{aligned}$$

and there is a unique topology on X and space of functions $\mathcal{O}_X$ such that U_i is open in X and θ_i is a regular isomorphism for all $i = 0...n$.

Denote this by $\mathbb{P}_k^n$. With this structure $\mathbb{P}_k^n$ is an irreducible, separated algebraic variety of dimension n.

Further for any homogenous polynomial $F \in k[X_0,...,X_n]$ of degree d the expression

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$$(x_0 : ... : x_n) \to \frac{F(x_0, ..., x_n)}{x_i^d}$$

is a well-defined regular function on U_i. In particular on U_i we may define the ***homogenous coordinate functions*** $\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}$.

Proof. It is straightforward to show that θ_i is a bijection. Define the cocycle maps

$$\theta_{ij} : \theta_i(U_i \cap U_j) \to \theta_j(U_i \cap U_j)$$

by $\theta_{ij} = \theta_j \circ \theta_i^{-1}$. We show that $\theta_i(U_i \cap U_j)$ is open in $\mathbb{A}_k^n\big(\overline{k}\big)$ and the maps θ_{ij} are regular. First observe that

$$\theta_i(U_i \cap U_j) = \begin{cases} D(Z_j) & i < j \\ D(Z_{j+1}) & j < i \end{cases}$$

Then the cocycle map is given pointwise by

$$\theta_{ij} : D(Z_j) \to D(Z_{i+1})$$
$$(z_1, ..., z_n) \to \left(\frac{z_1}{z_j}, ..., \frac{z_i}{z_j}, \frac{1}{z_j}, ..., \frac{z_n}{z_j}\right)$$

Using (6.2.45) and (6.1.39) we may show that θ_{ij} is a regular morphism of abstract varieties. Therefore the structure on X is uniquely determined by (6.2.38).

By construction U_i is affine and $U_i \cap U_j = D\left(\frac{X_j}{X_i}\right)$ where $\frac{X_j}{X_i} \in \mathcal{O}_X(U_i)$. Therefore $U_i \cap U_j$ is affine by (6.2.21) and there is a canonical isomorphism

$$\mathcal{O}_{U_i}(U_i)_{\frac{X_j}{X_i}} \xrightarrow{\sim} \mathcal{O}_{U_i}(U_i \cap U_j)$$

As $\frac{X_i}{X_j} \in \mathcal{O}_{U_j}(U_j)$ satisfies $\frac{X_i}{X_j}|_{U_i \cap U_j} = \left(\frac{X_j}{X_i}|_{U_i \cap U_j}\right)^{-1}$ we find the canonical map

$$\mathcal{O}_{U_i}(U_i) \otimes \mathcal{O}_{U_j}(U_j) \to \mathcal{O}_{U_i}(U_i \cap U_j)$$

is surjective. Therefore X is separated by (6.2.69).

By (6.1.21) the affine space $\mathbb{A}_k^n$ is irreducible whence so is U_i and $U_i \cap U_j$ (4.1.70). Therefore by (4.1.74) $\mathbb{P}_k^n$ is irreducible and has dimension n by (4.1.88). ∎

Proposition 6.2.94 (Riemann Sphere)
There is a bijection

$$\mathbb{P}_k^1 \xrightarrow{\sim} \mathbb{A}_k^1 \cup \{\infty\}$$

which sends $(x : 1)$ to x and $(1 : 0)$ to ∞. This gives $\mathbb{A}_k^1 \cup \{\infty\}$ the structure of a variety under which $\mathbb{A}_k^1$ is an open subvariety and x^{-1} is a regular automorphism interchanging 0 and ∞.

Consider $x = \frac{X_1}{X_0} \in \mathcal{O}_{\mathbb{P}_k^1}(U_0)$ as an element of the function field $k(\mathbb{P}_k^1)$ then x is transcendental and $k(\mathbb{P}_k^1) = k(x)$.

Proof. Evidently x is transcendental because $\mathcal{O}(U_0) = k[x]$ (...). Let $f \in k(\mathbb{P}_k^1)$ and suppose $\mathrm{dom}(f) \cap U_0 \neq \emptyset$. There is some $g(x) \in \mathcal{O}(U_0)$ such that $D(g(x)) \subset \mathrm{dom}(f)$. Then by (...) $\mathcal{O}(D(g)) = k[x]_{g(x)}$ and so $f|_{D(g(x))} = h(x)/g(x)^r$ for some polynomial $h \in k[X]$ and integer $r > 0$. Therefore $f \in k(x)$. Similarly if $\mathrm{dom}(f) \cap U_1 \neq \emptyset$ then $f \in k(x^{-1}) = k(x)$ as required, where $x^{-1} = \frac{X_0}{X_1} \in \mathcal{O}_{\mathbb{P}_k^1}(U_1)$. ∎

We show later that all projective closed subvarieties are complete, but here we demonstrate a simpler case more directly.

Proposition 6.2.95
$\mathbb{P}_k^1$ is complete.

Proof. Let $Z \subset \mathbb{P}_k^1$ be an irreducible closed subvariety. Then by (4.1.83) (6.2.56) it is either the (closure of) a single point or the whole space. In the former case the function field is algebraic over k and has no non-trivial valuation rings. Therefore we may assume wlog that $Z = \mathbb{P}_k^1$ and consider a valuation ring $R \subset k(\mathbb{P}_k^1) = k(x)$ where $x = \frac{X_1}{X_0}$. By (6.2.118) R is equal to either $k[x]_{p(x)}$ for some irreducible polynomial or $k\big[x^{-1}\big]_{x^{-1}}$. Let α be a root of $p(x)$ in $\overline{k}$, then this corresponds to the point $(1 : \alpha)$, or in the latter case just $(0 : 1)$. ∎

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6.2.17 Affine and Finite Morphisms

Definition 6.2.96
*Let $\phi : X \to Y$ be a regular morphism of varieties. We say ϕ is **affine** if for every affine open $V \subset Y$, $\phi^{-1}(V)$ is affine.*

*We say in addition that ϕ is **finite** if the ring map $\mathcal{O}_Y(V) \to \mathcal{O}_X(\phi^{-1}(V))$ is finite (and therefore integral ...).*

Other than the definition we do not require this section in the sequel. But it may be convenient for other applications to show that it's sufficient to demonstrate a morphism is affine (resp. finite) only over a single affine cover.

Lemma 6.2.97
Let X be a variety with U, V open affine subsets. Then there exists $g_1, ..., g_r \in \mathcal{O}_X(U)$ and $g'_1, ..., g'_r \in \mathcal{O}_X(V)$ such that

a) $W_i := D(g_i) = D(g'_i)$ for $i = 1...r$
b) $U \cap V = \bigcup_{i=1}^r W_i$

Proof. Given $x \in U \cap V$, then as $U \cap V$ is open in U by (6.2.10) we may find $f_x \in \mathcal{O}_X(U)$ such that $x \in D(f_x) \subset U \cap V$. Similarly we may find $f'_x \in \mathcal{O}_X(V)$ with $x \in D(f'_x) \subset D(f_x)$. By (6.2.21).c) there is a canonical isomorphism $\mathcal{O}_X(U)_{f_x} \xrightarrow{\sim} \mathcal{O}_X(D(f_x))$. Therefore $f'_x|_{D(f_x)} = \frac{g}{(f_x)^r}$ for some $g \in \mathcal{O}_X(U)$. Observe that $D(f'_x) = D(g) \cap D(f_x) = D(gf_x)$. This gives the required pair of sections for an arbitrary $x \in U \cap V$. Clearly these open sets cover $U \cap V$, and as $U \cap V$ is quasi-compact (6.2.7) we are done. ∎

Proposition 6.2.98
Let $\phi : X \to Y$ be a regular morphism of varieties, and suppose there is an open cover $Y = \bigcup_{i=1}^n V_i$ for which V_i and $\phi^{-1}(V_i)$ are affine. Then ϕ is affine.

If in addition $\mathcal{O}_Y(V_i) \to \mathcal{O}_X(\phi^{-1}(V_i))$ is finite for all $i = 1...n$ then ϕ is finite.

Proof. By assumption we have affine open sets $U_i := \phi^{-1}(V_i)$ for $i = 1...n$. Let $V \subset Y$ be an open affine set, then by (6.2.97) for each i we have an open cover

$$V \cap V_i = \bigcup_{j=1}^{n_i} W_{ij}$$

with $W_{ij} := D(f_{ij}) = D(f'_{ij})$ for $f'_{ij} \in \mathcal{O}_Y(V_i)$ and $f_{ij} \in \mathcal{O}_Y(V)$. Evidently

$$V = \bigcup_{i=1}^{n} \bigcup_{j=1}^{n_i} D(f_{ij}) \quad (*)$$

and so by (6.2.26) the f_{ij} generate $\mathcal{O}_Y(V)$, and consequentally $g_{ij} := \phi^\sharp(f_{ij})$ generate $\mathcal{O}_X(U)$ (as both ideals contain 1). Further $\phi^{-1}(W_{ij}) = \phi^{-1}(D(f'_{ij})) = D(\phi^\sharp(f'_{ij})) \subset U_i$ is affine (6.2.21).c). Therefore from (6.2.28) U is affine, because we also have the representation $\phi^{-1}(W_{ij}) = \phi^{-1}(D(f_{ij})) = D(g_{ij})$.

Suppose ϕ is finite then by definition this means the ring maps $\mathcal{O}_Y(V_i) \to \mathcal{O}_X(U_i)$ are finite. By (3.5.48).a) this means the localised maps are finite

$$\begin{array}{ccc}
\mathcal{O}_Y(V_i)_{f'_{ij}} & \longrightarrow & \mathcal{O}_X(U_i)_{g'_{ij}} \\
\downarrow \sim & & \downarrow \sim \\
\mathcal{O}_Y(W_{ij}) & \xrightarrow{\phi^\sharp} & \mathcal{O}_X(\phi^{-1}(W_{ij})) \\
\uparrow \sim & & \uparrow \sim \\
\mathcal{O}_Y(V)_{f_{ij}} & \dashrightarrow & \mathcal{O}_X(U)_{g_{ij}}
\end{array}$$

Using equation $(*)$ and (6.2.26) we see that f_{ij} generate the unit ideal of $\mathcal{O}_Y(V)$. Therefore by (3.5.48).b) $\phi^\sharp_V$ is finite. ∎

6.2.18 Normal Varieties

Definition 6.2.99
*Let A be an integral domain. We say that the ring A is **normal** if it is integrally closed in its field of fractions.*

Recall that A factorial $\implies$ A is normal (3.23.12).

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Proposition 6.2.100
The polynomial ring $A[X_1, ..., X_n]$ is normal for every unique factorisation domain A.

Proof. See (3.23.12) and (3.16.46). ∎

Proposition 6.2.101 (Normal Variety)
Let X be an integral variety. Then the following are equivalent

a) $\mathcal{O}_X(U)$ is normal for every open subset $U \subset X$
b) $\mathcal{O}_{X,W}$ is normal for every irreducible subset $W \subset X$
c) $\mathcal{O}_{X,x}$ is normal for every point $x \in X$

*In this case we say that X is a **normal variety**. If X is affine then this is equivalent to $\mathcal{O}_X(X)$ being normal.*

Proof. *a)* $\implies$ *b)* Let $U \subset X$ be any affine open subset such that $U \cap W \neq \emptyset$, then by (...) $\mathcal{O}_{X,W} \xrightarrow{\sim} \mathcal{O}_{U,W\cap U} \xrightarrow{\sim} \mathcal{O}_X(U)_{\mathfrak{m}_{U,W\cap U}}$, and the result follows from (3.23.10).

b) $\implies$ *c)* is straightforward because $\mathcal{O}_{X,x} = \mathcal{O}_{X,\{x\}}$.

c) $\implies$ *a)* We first show this for the case that U is affine. By (...) $\mathcal{O}_{X,x} \xrightarrow{\sim} \mathcal{O}_{U,x} \xrightarrow{\sim} \mathcal{O}_X(U)_{\mathfrak{m}_{U,x}}$ is normal for every $x \in U$. Therefore by (6.2.21).b) and (3.23.10) $\mathcal{O}_X(U)$ is normal.

Now let U be an arbitrary open set and $V \subset U$ affine. Then by (6.2.15) we have inclusions

$$\mathcal{O}_X(V) \hookrightarrow \mathcal{O}_X(U) \hookrightarrow \operatorname{Frac}(\mathcal{O}_X(U)) \hookrightarrow k(X)$$

and $k(X)$ is a fraction field for $\mathcal{O}_X(V)$. If $\alpha \in \operatorname{Frac}(\mathcal{O}_X(U))$ is integral over $\mathcal{O}_X(U)$, then it is a-fortiori integral over $\mathcal{O}_X(V)$. By the affine case then $\alpha \in \mathcal{O}_X(V)$ and we conclude that $\mathcal{O}_X(U)$ is normal as required.

For the case X is affine we may use (3.23.10). ∎

Corollary 6.2.102
The standard affine space $\mathbb{A}^n_k\left(\overline{k}\right)$ is normal.

Proof. The coordinate ring is a UFD (3.16.46) and therefore normal (3.23.12). ∎

Proposition 6.2.103
Let X be an integral variety with open cover $X = \bigcup_{i=1}^n U_i$. Then X is normal iff U_i are normal.

Proof. Straightforward, since normality is a local condition. ∎

6.2.19 Weil Divisors

The notion of regular curves can be generalised in one direction.

Definition 6.2.104 (Prime Divisors of an Integral Variety)
*Let X be an integral, separated variety. We say it is **regular in codimension one** if for every irreducible closed subset $Y \subset X$ for which $\operatorname{codim}(Y, X) = 1$ the local ring $\mathcal{O}_{X,Y}$ is regular (equivalently integrally closed or a DVR (3.30.2)).*

*In this case, we say an irreducible closed subset of codimension 1 is a **prime divisor**.*

We denote by $\operatorname{Div} X$ the free abelian group generated by the prime divisors. We write a prime divisor D by a formal sum

$$D = \sum_{Y \subset X} n_Y Y$$

*where $n_Y \in \mathbb{Z}$ and all but finitely many are zero. We define the **degree** of a divisor D by*

$$\deg D = \sum_{Y \subset X} n_Y \left[k\left(\mathfrak{m}_{X,Y}\right) : k\right]$$

Proposition 6.2.105
An integral, separated variety X is regular in codimension one for either of the following special cases

a) X is normal
b) X is non-singular and k is perfect.

Proof. See (3.30.2) and (6.2.82). ∎

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Proposition 6.2.106 (Valuation Associated to Prime Divisor)
Let X be a variety satisfying (6.2.104) and $Y \subset X$ a prime divisor. Then $\mathcal{O}_{X,Y}$ is a discrete valuation ring.

We denote by $v_Y : k(X)^\star \to \mathbb{Z}$ the corresponding valuation.

Proof. By (6.2.55) $\dim \mathcal{O}_{X,Y} = \operatorname{codim}(Y, X) = 1$ and is integral, so the result follows from (3.30.2). ∎

Proposition 6.2.107 (Prime Divisors on a Curve)
Let X be a regular curve. Then X satisfies (6.2.104) and the prime divisors are precisely irreducible closed subsets of the form $\overline{\{x\}}$. These are in bijective correspondence with the valuation rings of $k(X)$, via the map $x \to \mathcal{O}_{X,x}$.

Proof. By the codimension formula (6.2.55) the irreducible closed subsets of codimension 1 are precisely those of dimension 0. Therefore the result follows from (6.2.125). ∎

Proposition 6.2.108
Let X be a variety satisfying (6.2.104) and $f \in k(X)^\star$ a rational function. Then $v_Y(f) = 0$ for all but finitely many prime divisors Y.

Proof. By definition f is regular on some open subset U of X, which we may assume to be affine. Define $Z \coloneqq X \setminus U$. Then by (...) $Z = Z_1 \cup ... \cup Z_n$ is a decomposition into irreducible components. By (4.1.83).f) $\operatorname{codim}(Z, X) > 0$ whence $\operatorname{codim}(Z_i, X) > 0$ for all $i = 1...n$. Suppose $Y \subset Z$ is a prime divisor then by definition $\operatorname{codim}(Y, X) = 1$ and by (4.1.66) $Y \subset Z_i$ for some i. Therefore by (4.1.83).c) $\operatorname{codim}(Y, Z_i) \leq \operatorname{codim}(Y, X) - \operatorname{codim}(Z_i, X) \leq 0$ and therefore by (4.1.83).f) $Y = Z_i$.

We conclude all but finitely many Y intersect U, so we may without loss of generality assume $Y \cap U \neq \emptyset$. Evidently U also satisfies (6.2.104) and by (4.1.79) there is a bijection between prime divisors of X meeting U, and prime divisors of U.

Therefore we may assume that X is affine and $0 \neq f \in \mathcal{O}_X(X)$. Then $v_Y(f) > 0 \iff f \in \mathfrak{M}_{X,Y} \iff f \in I(Y) \iff Y \subset V(f)$. We claim $V(f)$ is a proper subset of X (for $V(f) = X \iff \sqrt{(0)} = \sqrt{f} \iff f = 0$). Therefore $\operatorname{codim}(V(f), X) > 0$ and

$$\operatorname{codim}(Y, V(f)) \leq \operatorname{codim}(Y, X) - \operatorname{codim}(V(f), X) \leq 0$$

so Y is an irreducible component of $V(f)$, of which there are only finitely many (4.1.66). ∎

Definition 6.2.109 (Principal Divisor)
Let X be a variety satisfying (6.2.104) then define the homomorphism

$$\begin{aligned} k(X)^\star &\longrightarrow \operatorname{Div}(X) \\ f &\longrightarrow (f) \end{aligned}$$

where we define a formal sum over all prime divisors

$$(f) \coloneqq \sum_{Y \subset X} v_Y(f) Y.$$

*The co-kernel of this map is known as the **divisor class group** of X, and denoted $\operatorname{Cl}(X)$. We say that two divisors are equivalent $D \sim D'$ if they become equal as elements of the divisor class group. Equivalently if $D - D' = (f)$ for some $f \in k(X)^\star$.*

6.2.20 Algebraic Function Fields

We develop the basic theory of **algebraic function fields** of one variable, the field-theoretic counterpart to a regular algebraic curve.

Proposition 6.2.110
Let K/k be a field extension. Then the following are equivalent

a) K/k is finitely generated with transcendence degree 1.
b) There exists a transcendental element $x \in K/k$ such that $K/k(x)$ is finite.

*In this case we say that K/k is an **algebraic function field**. We typically assume that k is perfect (either a number field or a finite field).*

Proof. *a)* $\implies$ *b)* By (3.18.139) a transcendence base consists of a single element x. By assumption $K/k(x)$ is algebraic and finitely generated, hence also finite (3.18.44).

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b) $\Longrightarrow$ *a)* Evidently $\{x\}$ is a transcendence base. By assumption $K/k(x)$ is finite and therefore certainly finitely generated, whence so is K/k. ∎

Definition 6.2.111 (Algebraic Function Field)
*Let K/k be an algebraic function field. Denote by $\tilde{k}$ the **field of constants**, the relative algebraic closure of k in K:*

$$\tilde{k} := \{x \in K \mid x \text{ algebraic over } k\}$$

*(a subfield of K by (3.18.45)). By (3.18.46), $K/\tilde{k}$ is relatively algebraically closed. We say that K/k is **geometrically integral** if $k = \tilde{k}$ (c.f. (3.35.41)).*

Definition 6.2.112 (Place of an Algebraic Function Field)
*Let K/k be an algebraic function field. By a **place** of K/k we refer to a valuation ring $\mathcal{O}_P$ such that $k \subsetneq \mathcal{O}_P \subsetneq K$. By (3.24.2) this is a local ring with unique maximal ideal $\mathfrak{m}_P$.*

From (3.3.13) we verify immediately that $k \cap \mathfrak{m}_P = (0)$.

*For every $x \in K$ we say that x is **regular** at a place P if $x \in \mathcal{O}_P$.*

Proposition 6.2.113
Let K/k be a function field and $k \subsetneq \mathcal{O}_P \subsetneq K$ a valuation ring. Then $\tilde{k} \subset \mathcal{O}_P$ and $\tilde{k} \cap \mathfrak{m}_P = (0)$

Proof. By definition $\tilde{k}$ is the integral closure of k in K, so $\tilde{k} \subset \mathcal{O}_P$ by (3.24.14). Suppose $0 \neq x \in \tilde{k}$ then x is invertible in $\mathcal{O}_P$ and therefore x is not contained in any proper ideal, and in particular $x \notin \mathfrak{m}_P$. ∎

Proposition 6.2.114 (Residue Degree of a Place)
Let K/k be an algebraic function field and $f \in K \setminus \tilde{k}$. Then

a) $K/k(f)$ is finite.
b) If $\mathcal{O}_P$ is a place with $f \in \mathcal{O}_P$, then $\mathfrak{p} := \mathfrak{m}_P \cap k[f]$ is a maximal ideal of $k[f]$, the field $k(\mathfrak{p}) := k[f]/\mathfrak{p}$ is finite over k, and

$$[k(\mathfrak{m}_P) : k(\mathfrak{p})] \leq [K : k(f)]$$

c) Consequently the residue field $k(\mathfrak{m}_P)$ of every place P of K/k, whether or not it contains f, is a finite extension of k.

Proof. a) As f is not algebraic over k and $\operatorname{trdeg}(K/k) = 1$ (6.2.110), f is a transcendence base (3.18.139), so $K/k(f)$ is algebraic and finitely generated, hence finite (3.18.44).

b) The ring $k[f]$ is a polynomial ring, hence a PID (3.15.4), and $\mathfrak{p}$ is a prime ideal of it contained in $\mathfrak{m}_P$. Suppose $\mathfrak{p} = 0$. Then every non-zero element of $k[f]$ is a unit of $\mathcal{O}_P$, so $k(f) \subset \mathcal{O}_P$. Every element of K is algebraic over $k(f)$ by a), hence integral over $k(f)$, and $\mathcal{O}_P$ is integrally closed (3.24.2), so $K = \mathcal{O}_P$, a contradiction. So $\mathfrak{p}$ is a non-zero prime ideal of a PID, hence maximal, and $\mathfrak{p} = (g(f))$ for an irreducible polynomial g. Thus $k(\mathfrak{p}) \cong k[X]/(g)$ is finite over k (3.18.40). Further $k[f]_{\mathfrak{p}}$ is a discrete valuation ring (3.30.4), in particular a PID, and $\mathcal{O}_P$ is an overring of $k[f]$ with quotient field K containing the proper ideal $\mathfrak{m}_P \supsetneq \mathfrak{p}$. So (3.31.18) applied to $k[f] \subset \mathcal{O}_P$ gives the inequality.

c) As $\mathcal{O}_P \neq K$ is not a field, there is $0 \neq x \in \mathfrak{m}_P$. Then $x \notin \tilde{k}$ by (6.2.113), so b) applies to x, and $[k(\mathfrak{m}_P) : k] = [k(\mathfrak{m}_P) : k(\mathfrak{p})][k(\mathfrak{p}) : k]$ is finite. ∎

In order to use the theory of Dedekind domains and affine algebras we establish a connection between valuation rings and a certain "affine" subring.

Proposition 6.2.115
Let K/k be an algebraic function field and $f \in K \setminus \tilde{k}$ transcendental. Then the integral closure $\overline{k[f]}$ in K is a Dedekind domain with quotient field K and a finite $k[f]$-module. There is a correspondence

$$\left\{\mathfrak{B} \lhd \overline{k[f]} \;\middle|\; \mathfrak{B} \text{ non-zero prime}\right\} \longleftrightarrow \{\mathcal{O}_P \subsetneq K \mid f \in \mathcal{O}_P\}$$

$$\mathfrak{B} \longrightarrow \overline{k[f]}_{\mathfrak{B}}$$

$$\mathfrak{m}_P \cap \overline{k[f]} \longleftarrow \mathcal{O}_P$$

In particular every such $\mathcal{O}_P$ is a discrete valuation ring, and $k(\mathfrak{m}_P) = k(\mathfrak{B})$. Moreover every place P of K/k (whether or not $f \in \mathcal{O}_P$) is a discrete valuation ring, so we may associate to it a discrete valuation

$$v_P : K \longrightarrow \mathbb{Z} \cup \{\infty\}$$

which is zero on k and $v_P(f) < 0 \Longleftrightarrow f \notin \mathcal{O}_P \Longleftrightarrow f^{-1} \in \mathfrak{m}_P$.

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Proof. As $f \in K \setminus \tilde{k}$ is transcendental over k, $k[f]$ is a polynomial ring and so a PID (3.15.4), hence an affine Dedekind domain (3.31.2), which has Krull dimension 1 (3.31.1). Also $K/k(f)$ is finite (6.2.114).a). By (3.34.61) $B := \overline{k[f]}^K$ is therefore a Dedekind domain and finite $k[f]$-module, with quotient field K as $K/k(f)$ is algebraic.

Let P be a place with $f \in \mathcal{O}_P$. As $\mathcal{O}_P$ is a valuation ring, it is integrally closed (3.24.2) and as it contains $k[f]$, also contains B. By (3.31.4), $\mathfrak{B} := \mathfrak{m}_P \cap B$ is then a non-zero prime of B with $\mathcal{O}_P = B_{\mathfrak{B}}$ and $\mathfrak{m}_P = \mathfrak{B}B_{\mathfrak{B}}$; conversely each $B_{\mathfrak{B}}$ is a place, being a discrete valuation ring (hence a valuation ring) with $k \subsetneq B_{\mathfrak{B}} \subsetneq K$ as $f \in B_{\mathfrak{B}} \setminus k$. As $\mathfrak{B}$ is maximal, $B/\mathfrak{B}$ is a field, so $k(\mathfrak{m}_P) = B_{\mathfrak{B}}/\mathfrak{B}B_{\mathfrak{B}} \cong \operatorname{Frac}(B/\mathfrak{B}) = k(\mathfrak{B})$ (3.5.43).

This covers every place P with $f \in \mathcal{O}_P$. For a place P with $f \notin \mathcal{O}_P$ instead, $f^{-1} \in \mathfrak{m}_P \subset \mathcal{O}_P$ (3.24.2), and f^{-1} is transcendental too (else $f = (f^{-1})^{-1}$ would be algebraic); applying the above with f^{-1} in place of f shows $\mathcal{O}_P$ is again a discrete valuation ring. So every place P of K/k is a discrete valuation ring, giving the associated valuation v_P. It vanishes on k, as non-zero elements of k are units of $\mathcal{O}_P$, and $v_P(f) < 0 \iff f \notin \mathcal{O}_P \iff f^{-1} \in \mathfrak{m}_P$ by (3.24.2). ∎

Corollary 6.2.116
Let K/k be an algebraic function field. Then the field of constants $\tilde{k}/k$ has finite degree.

Proof. For any place P we have an embedding $\tilde{k}/k \hookrightarrow k(\mathfrak{m}_P)/k$ which is finite (6.2.114).c). ∎

We prove a number of facts in analogy with algebraic curves.

Lemma 6.2.117 (Every function has a zero)
Let K/k be an algebraic function field and $0 \neq f \in K$. If $f \in \tilde{k}$ (i.e. f is algebraic), then $v_P(f) = 0$ for every place P, so f has no zeros or poles; otherwise (f transcendental) f has between 1 and $[K : k(f)]$ zeros (resp. poles).

Proof. If $f \in \tilde{k}$, then $f \neq 0$ gives $f \in \tilde{k} \setminus \{0\}$, so $f \notin \mathfrak{m}_P$ for every place P (6.2.113), that is $v_P(f) = 0$.

Otherwise, clearly $k[f]$ is a polynomial ring with maximal ideal (f). Therefore by (3.24.12) there is a valuation ring $\mathcal{O}_P \supset k[f]$ such that $\mathfrak{m}_P \cap k[f] = (f)$, which means f has at least one zero.

Recall by (6.2.115) that zeros of f correspond to the maximal ideals of $\overline{k[f]}$ containing f. These are precisely the prime divisors of (f) (see (3.31.13)), so the bound follows from (3.31.20).

The statement about poles of f follows immediately because these are precisely zeros of f^{-1} and $k(f) = k(f^{-1})$. ∎

Proposition 6.2.118 (Places of the Rational Function Field)
Let $K = k(x)$ with x transcendental over k. Then the places of K/k, that is the valuation rings $k \subsetneq \mathcal{O} \subsetneq K$, are exactly the following pairwise distinct discrete valuation rings

a) $\mathcal{O}_{p(x)} := k[x]_{(p(x))}$ where $p \in k[X]$ is an irreducible monic polynomial, with uniformiser $p(x)$ and residue field $k[X]/(p)$, and
b) $\mathcal{O}_\infty := \left\{ \frac{f(x)}{g(x)} \;\middle|\; g \neq 0, \deg f \leq \deg g \right\}$, with uniformiser x^{-1} and residue field k.

If k is algebraically closed then the irreducible monic polynomials are the $X - \theta$ with $\theta \in k$, so every place is rational, with uniformiser $x - \theta$ or x^{-1}.

Proof. Let $t := x^{-1}$, which is again transcendental over k with $k(t) = K$.

The listed rings are discrete valuation rings. As $k[x] \cong k[X]$ is a PID (3.15.4), $\mathcal{O}_{p(x)}$ is a discrete valuation ring (3.30.4), with maximal ideal $p(x)\mathcal{O}_{p(x)}$ and residue field $k[x]/(p(x))$ (3.5.27).

Next $\mathcal{O}_\infty = k[t]_{(t)}$: if $f, g \in k[X]$ with $g(0) \neq 0$ and $N := \max(\deg f, \deg g)$ then $f(t)/g(t) = F(x)/G(x)$ with $F(x) := x^N f(1/x)$, $G(x) := x^N g(1/x)$, $\deg F \leq N = \deg G$; conversely $F(x)/G(x)$ with $\deg F \leq \deg G =: N$ equals $f(t)/g(t)$ with $f(t) := t^N F(1/t)$ and $g(t) := t^N G(1/t)$, where $g(0)$ is the leading coefficient of G. So $\mathcal{O}_\infty$ is a discrete valuation ring with uniformiser t and residue field $k[t]/(t) = k$, as above.

Every place is listed. Let $\mathcal{O}$ be a place with maximal ideal $\mathfrak{m}$. Suppose firstly that $x \in \mathcal{O}$. Then $k[x] \subset \mathcal{O}$ and the ideal $k[x] \cap \mathfrak{m} =: (p(x))$ is principal. There is an injective k-algebra homomorphism $k[x]/(p(x)) \hookrightarrow k(\mathfrak{m})$ whence $(p(x))$ is prime (3.3.59). As $k(\mathfrak{m})/k$ is algebraic (6.2.114).c) and x is transcendental, we must have $0 \neq p$, which we take monic, and therefore p is irreducible (3.16.11). Suppose $g(x) \notin (p(x))$ then $g(x) \notin \mathfrak{m} \implies g(x)^{-1} \in \mathcal{O}$ (3.24.2). Therefore $\mathcal{O}_{p(x)} \subset \mathcal{O}$, and as $\mathcal{O}_{p(x)}$ is a discrete valuation ring and $\mathcal{O} \neq K$ we conclude that they are equal (3.30.5).

Alternatively suppose that $x \notin \mathcal{O}$ then by (3.24.2) $t = x^{-1} \in \mathfrak{m} \subset \mathcal{O}$. Applying the previous part with t in place of x gives $\mathcal{O} = \mathcal{O}_{p(t)}$ where $\mathfrak{m} \cap k[t] = (p(t))$. As $t \in \mathfrak{m}$ we conclude that $p(t) = t$, and $\mathcal{O} = k[t]_{(t)} = \mathcal{O}_\infty$.

Distinctness. $x \in \mathcal{O}_{p(x)}$ but $x \notin \mathcal{O}_\infty$, as $\deg x > \deg 1$. For distinct monic irreducible $p \neq p'$, $p \notin (p')$, so $p(x)$ is a unit of $\mathcal{O}_{p'(x)}$ but not of $\mathcal{O}_{p(x)}$.

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Finally if k is algebraically closed the irreducible monic polynomials are the $X-\theta$ (3.18.58), with residue field $k[X]/(X-\theta)=k$. ∎

Lemma 6.2.119 (Restricting a Place is Proper)
Let L/k' be an algebraic function field, P' a place of L/k', $k\subseteq k'$ with k'/k algebraic, and $K\subseteq L$ a subfield with $\mathrm{trdeg}_k(K)\geq 1$ (e.g. K/k itself an algebraic function field). Then $\mathcal{O}_{P'}\cap K$ is a proper valuation ring of K (and therefore a place.)

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\mathcal{O}_{P'}\cap K$ is a valuation ring of K (3.24.3). Suppose it equals K, so $K\subset\mathcal{O}_{P'}$. Nonzero elements of K are units of $\mathcal{O}_{P'}$, hence not in $\mathfrak{m}_{P'}$, so $K\to\mathcal{O}_{P'}/\mathfrak{m}_{P'}=k'(\mathfrak{m}_{P'})$ is injective, and $\mathrm{trdeg}_k(K)\leq\mathrm{trdeg}_k(k'(\mathfrak{m}_{P'}))$ (3.18.136). But $k'(\mathfrak{m}_{P'})/k'$ is algebraic (6.2.114).c) and k'/k is algebraic, so the right-hand side is 0, contradicting $\mathrm{trdeg}_k(K)\geq 1$. ∎

Lemma 6.2.120 (Places Restrict Correctly)
Let K/k be an algebraic function field, P a place of K/k, and L/K a field extension. Let $k\subseteq k''\subset L$ with k''/k algebraic and L/k'' itself an algebraic function field, and P' a place of L/k'' with $\mathcal{O}_P\subset\mathcal{O}_{P'}$. Then $\mathcal{O}_{P'}\cap K=\mathcal{O}_P$; in particular $\mathfrak{m}_P\subset\mathfrak{m}_{P'}$, and $\pi_P\notin k''$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\mathcal{O}_{P'}\cap K$ is a valuation ring of K containing $\mathcal{O}_P$ (3.24.3), and proper (as K/k is an algebraic function field, so $\mathrm{trdeg}_k(K)=1\geq 1$) (6.2.119). By (3.30.5) ($\mathcal{O}_P$ being a DVR, (6.2.115)), a valuation ring of K containing the DVR $\mathcal{O}_P$ is either $\mathcal{O}_P$ itself or all of K; as $\mathcal{O}_{P'}\cap K\neq K$, this forces $\mathcal{O}_{P'}\cap K=\mathcal{O}_P$.

For $0\neq x\in\mathfrak{m}_P$ we have $x^{-1}\in K\setminus\mathcal{O}_P=K\setminus\mathcal{O}_{P'}$, so $x\in\mathfrak{m}_{P'}$ by (3.24.2): thus $\mathfrak{m}_P\subset\mathfrak{m}_{P'}$. In particular $\pi_P\in\mathfrak{m}_{P'}$ is a non-unit of $\mathcal{O}_{P'}$, so $\pi_P\notin k''$ (nonzero elements of the field $k''\subset\mathcal{O}_{P'}$ are units). ∎

6.2.21 Algebraic Curves

Definition 6.2.121
*We say an integral, separated algebraic variety of dimension 1 is an **algebraic curve** or just a **curve**.*

Proposition 6.2.122 (Characterisation Regular Point on a curve)
Let X be an algebraic curve and $x\in X$. Then the following are equivalent

a) x is regular
b) $\mathcal{O}_{X,x}$ is a discrete valuation ring
c) $\mathcal{O}_{X,x}$ is an integrally closed domain

and when k is perfect this is equivalent to x being non-singular.

In particular X is regular if and only if it is normal.

Proof. Follows exactly the same lines as (6.1.100). ∎

Example 6.2.123
Both $\mathbb{A}^1_k$ and $\mathbb{P}^1_k$ are regular algebraic curves.

Proposition 6.2.124 (Curve has Cofinite Topology)
Let X be an algebraic curve (or just an algebraic variety of dimension 1) then X_0 has the cofinite topology.

Proof. Follows from (4.1.92). ∎

Proposition 6.2.125 (Points on Complete Regular Curve = Valuation Rings on Function Field)
Let X be a complete regular algebraic curve. Then there is a bijection

$$\{W\subset X\mid \dim W=0 \text{ closed and irreducible}\}\longleftrightarrow X_0\longleftrightarrow\{k\subset R\subsetneq k(X)\mid R \text{ a valuation ring}\}$$

Further the corresponding valuation rings are all discrete valuation rings.

For an affine open subset $U\subset X$ this restricts to a bijection

$$U_0\longleftrightarrow\{\mathcal{O}_X(U)\subset R\subsetneq k(X)\mid R \text{ a valuation ring}\}$$

where $\mathcal{O}_X(U)$ is identified as a subring of $k(X)$ (6.2.18).

Proof. The first map is (4.1.94) and the second map is well-defined by (6.2.122) and injective by (6.2.20).

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Let $R \subset k(X)$ be a valuation ring then by definition of completeness there is a point $x \in X$ such that $\mathcal{O}_{X,x} \subset R$. By (6.2.55) we have $\dim \mathcal{O}_{X,x} = \dim X = 1$. By (6.1.92) and (6.1.91) $1 = \dim \mathcal{O}_{X,x} \leq \dim R \leq \operatorname{trdeg}_k k(X) \overset{(6.2.54)}{=} \dim X = 1$. Therefore $\dim R = 1$ and by the same result $R = \mathcal{O}_{X,x}$. This shows that the map is surjective.

For the final statement we may use the affine case (6.1.101) and the commutative diagram

$$\begin{array}{ccc} \mathcal{O}_X(U) & \hookrightarrow & k(X) \\ \| & & \downarrow \sim \\ \mathcal{O}_U(U) & \hookrightarrow & k(U) \end{array}$$

■

Proposition 6.2.126
Let X be a regular algebraic curve and Y a complete variety. Then the rational maps are precisely the regular maps. In otherwords the canonical map

$$\operatorname{Mor}(X,Y) \xrightarrow{\sim} \operatorname{Rat}(X,Y)$$

is a bijection. Further this restricts to dominant maps.

Proof. Clearly the map is well-defined. Suppose that two regular maps ϕ, ψ are equal as rational maps. Then by definition they agree on a dense open subset. By definition Y is separated so by (6.2.63) the equaliser is a closed subset, and therefore the whole of Y. This means the map is injective.

Consider a regular map $\phi : U \to Y$ for U a (dense) open subset of X. Then $Z \coloneqq X \setminus U$ is a proper closed subset of X and therefore $\dim Z = 0$ by (4.1.84). By (6.2.12) we have $Z = \overline{\{z_1\}} \cup ... \cup \overline{\{z_n\}}$, and we may apply (6.2.90) finitely many times to lift to a regular map on an open set containing U and $z_1, ..., z_n$. By (4.1.41) this is the whole of X. Therefore the given map is surjective. ■

Proposition 6.2.127 (Image of Complete Curve either Constant or Surjective)
Let X, Y be algebraic curves with X complete and $\phi : X \to Y$ a regular morphism. Then ϕ is either surjective or has image of the form $\overline{\{x\}}$ (which we call constant).

Proof. By (6.2.89) ϕ is universally closed, so by (6.2.84) $\phi(X)$ is an irreducible closed subvariety of Y. Evidently $\dim \phi(X) = 0$ or 1. Then we may use (6.2.56) or (4.1.84). ■

Corollary 6.2.128 (Equivalence between nice curves and function fields)
Let X, Y be complete, regular algebraic curves. Then there are bijections

$$\{\phi : X \to Y \text{ non-constant}\} \leftrightarrow \{\phi : X \longrightarrow Y \text{ dominant}\} \leftrightarrow \{\phi_\star : k(Y) \hookrightarrow k(X)\}$$

All such morphisms are surjective and the degree

$$\deg(\phi) \coloneqq [k(X) : \phi_\star(k(Y))]$$

is always finite.

Finally the fibres are finite (i.e. ϕ is "quasi-finite").

Proof. This follows from (6.2.127), (6.2.126) and (6.2.76).

The fibres are finite because $\phi^{-1}(\{y\}) \subset \phi^{-1}\left(\overline{\{y\}}\right)$. By assumption this is a proper closed subset of X and therefore has dimension 0. So we are done by (6.2.56). ■

Proposition 6.2.129
Let X be a regular algebraic curve. Then there is a bijection

$$k(X)^\star \leftrightarrow \{\phi : X \to \mathbb{P}^1_k \text{ non-constant}\}$$

More precisely given $f \in k(X)^\star$, then for $x \in X$ the function

$$\phi_f(x) = \begin{cases} (f(x) : 1) & f \in \mathcal{O}_{X,x} \\ (1 : f^{-1}(x)) & f^{-1} \in \mathcal{O}_{X,x} \end{cases}$$

is well-defined and regular.

Proof. Recall (6.2.122) (6.2.15) that $\mathcal{O}_{X,x}$ is a (discrete) valuation ring with field of fractions $k(X)$.

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Define the open subsets $U \coloneqq \mathrm{dom}(f)$ and $V \coloneqq \mathrm{dom}(f^{-1})$. We claim that $X = U \cup V$. For $x \in \mathrm{dom}(f) \iff f \in \mathcal{O}_{X,x}$ and as $\mathcal{O}_{X,x}$ is a valuation ring for $k(X)$ then we have either $x \in U$ or $x \in V$.

Define $\phi : U \to \mathbb{P}^1_k$ by $\phi(x) = (f(x) : 1)$, and similarly $\psi : V \to \mathbb{P}^1_k$ by $\psi(x) = (1 : f^{-1}(x))$. By definition of projective structure sheaf (6.2.93) these are regular. Then for $x \in U \cap V$ we must have $f \notin \mathfrak{m}_{X,x}$ that is $f(x) \neq 0$ and $f^{-1}(x) = f(x)^{-1}$. This shows that $\phi|_{U\cap V} = \psi|_{U\cap V}$. Therefore they glue to a well-defined regular map $X \to \mathbb{P}^1_k$ as required.

We note that $\phi_f^{-1}(U_1) = \mathrm{dom}(f)$ where U_1 is the second affine piece, and we conclude the map is injective.

Conversely given a morphism $\phi : X \to \mathbb{P}^1_k$ then $U \coloneqq \phi^{-1}(U_1)$ is a non-empty open subset of X and $f : X_0/X_1 \circ \phi|_U : U \to \overline{k}$ is a regular function. Evidently $\phi|_U = \phi_f|_U$, and we conclude $\phi = \phi_f$ by (6.2.126). Therefore the map is surjective. ∎

Proposition 6.2.130 (Regular Morphisms of Complete Curves are Finite)
Let $\phi : X \to Y$ be a non-constant morphism of complete regular algebraic curves. Suppose $V \subset Y$ is an affine open subset, then

$$\mathcal{O}_X(\phi^{-1}(V)) = \overline{\phi_\star(\mathcal{O}_Y(V))}$$

where $\bar{\cdot}$ denotes the integral closure in $k(X)$.

Further ϕ is a finite (and therefore affine, and integral) morphism.

Proof. Define $A \coloneqq \mathcal{O}_Y(V)$ and $B \coloneqq \overline{\phi_\star(A)}$. Then

$$\begin{aligned} B &= \bigcap_{\phi_\star(A)\subset R} R && (3.24.14)\\ &= \bigcap_{A\subset\phi_\star^{-1}(\mathcal{O}_{X,x})} \mathcal{O}_{X,x} && (6.2.125)\\ &= \bigcap_{A\subset\mathcal{O}_{Y,\phi(x)}} \mathcal{O}_{X,x} && (6.2.18)\\ &= \bigcap_{\phi(x)\in V} \mathcal{O}_{X,x} && (6.2.125)\\ &= \mathcal{O}_X(\phi^{-1}(V)) && (6.2.73) \end{aligned}$$

By (6.2.54) $\dim V = \dim Y = 1$ and therefore $\dim \mathcal{O}_Y(V) = 1$ (6.1.50). By (3.34.61) B is a Dedekind domain and a finite A-module with quotient field $k(X)$.

Let $U \coloneqq \phi^{-1}(V)$ and observe that as $\mathcal{O}_{X,x}$ is integrally closed

$$B \subset \mathcal{O}_{X,x} \iff \phi_\star(A) \subset \mathcal{O}_{X,x} \iff A \subset \mathcal{O}_{Y,\phi(x)} \iff \phi(x) \in V \iff x \in U$$

where we have used the fact V is affine (6.2.125). This shows that there is a bijection between U_0 and the valuation rings of $k(X)$ containing $\mathcal{O}_X(U)$. It remains to show that U is affine.

Let Z denote an integral affine curve with coordinate ring (isomorphic to) $B = \mathcal{O}_X(U)$. By the Affine Mapping Property (6.2.22) (applicable to U without yet knowing it is affine, since only Z needs to be) there is a corresponding regular morphism

$$\psi : U \to Z$$

such that $\psi_Z^\sharp : \mathcal{O}_Z(Z) \to \mathcal{O}_U(U)$ is an isomorphism. Consider the commutative

$$\begin{array}{ccc} \mathcal{O}_Z(Z) & \xrightarrow[\sim]{\psi_Z^\sharp} & \mathcal{O}_U(U) \\ \downarrow & & \downarrow \\ k(Z) & \xrightarrow[\sim]{\psi_\star} & k(U) \end{array}$$

as $k(Z)$ is the field of fractions of $\mathcal{O}_Z(Z)$ and similarly for $\mathcal{O}_U(U)$ (6.2.15), we see that $\psi_\star$ is an isomorphism. In light of the correspondence of valuation rings to points we see that $\psi_0 : U_0 \to Z_0$ is a bijection. As these have the cofinite topology (6.2.124) then ψ_0 is necessarily a homeomorphism.

For $g \in \mathcal{O}_Z(Z)$ we have a commutative diagram

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$$\begin{array}{ccc}
\mathcal{O}_Z(Z) & \xrightarrow{\sim} & \mathcal{O}_U(U) \\
\downarrow & & \downarrow \\
\mathcal{O}_Z(D(g)) & \xrightarrow{\psi^\sharp_{D(g)}} & \mathcal{O}_U\big(D(\psi^\sharp(g))\big) \\
\uparrow \sim & & \uparrow \sim \\
\mathcal{O}_Z(Z)_g & \xrightarrow{\sim} & \mathcal{O}_U(U)_{\psi^\sharp(g)}
\end{array}$$

where the bottom isomorphisms follow from (6.2.27). This shows that $\psi^\sharp_{D(g)}$ is an isomorphism, and in particular (as the $D(g)$ form a basis of Z) that ψ is continuous. We claim that the morphism of stalks

$$\psi_x : \mathcal{O}_{Z,\psi(x)} \to \mathcal{O}_{U,x}$$

is an isomorphism for all $x \in U$. Suppose we have $(W, \sigma) \in \mathcal{O}_{U,x}$ and let $g \in \mathcal{O}_Z(Z)$ be such that $\psi(x) \in D(g) \subset \psi(W)$ (6.1.15). Then as $\psi^\sharp_{D(g)}$ is an isomorphism there exists $\tau \in \mathcal{O}_Z(D(g))$ such that $\psi^\sharp(\tau) = \sigma|_{\psi^{-1}(D(g))}$ which has the required image in $\mathcal{O}_{U,x}$. Similarly suppose that $(T, \tau) \in \mathcal{O}_{Z,\psi(x)}$ maps to zero. Then there exists some neighbourhood $x \in W \subset \psi^{-1}(T)$ such that $\psi^\sharp(\tau)|_W = 0$. We may choose $g \in \mathcal{O}_Z(Z)$ such that $D(g) \subset \psi(W)$. Then $0 = \psi^\sharp(\tau)|_{\psi^{-1}(D(g))} = \psi^\sharp\big(\tau|_{D(g)}\big)$ whence $\tau|_{D(g)} = 0$ which shows that (T, τ) is zero.

By (6.2.50) (applied to U and Z), together with ψ_0 already bijective and each stalk map ψ_x already an isomorphism, ψ is bijective. As ψ is a continuous bijection inducing the homeomorphism ψ_0 on Kolmogorov quotients, (4.1.38) shows ψ is a homeomorphism.

As $\psi^\sharp$ is an isomorphism on a basis of open sets, it is an isomorphism of sheaves. Therefore U is isomorphic to Z and affine as required. ∎

Definition 6.2.131 (Order of Zero / Poles)
Let X be a regular algebraic curve. For every point $x \in X$ there is a well-defined and unique discrete valuation

$$v_x : k(X)^\star \to \mathbb{Z}$$

*because $\mathcal{O}_{x,X}$ is a discrete valuation ring of $k(X)$ (6.2.125). For r a positive integer we say that $f \in k(X)^\star$ has a **zero** (resp. **pole**) of order r at x if $v_x(f) = r$ (resp. $v_x(f) = -r$).*

*As before define the **principal divisor***

$$(f) = \sum_{x \in X_0} v_x(f)x$$

on the free abelian group indexed by X_0.

Definition 6.2.132
*Let $\phi : X \to Y$ be a morphism of regular curves and $\phi_\star : k(Y) \hookrightarrow k(X)$ the correspond morphism of fields. Recall that for $x \in X$ that $\phi_\star$ restricts to a **local** homomorphism*

$$\mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{X,x}$$

*Therefore for $x \in X$ we may define the **ramification index** at x as follows*

$$e_x := v_x(\phi_\star(t)) \geq 1$$

*where t is a uniformiser for the discrete valuation ring $\mathcal{O}_{Y,\phi(x)}$. Similarly the **residue degree** is given by the degree of the field extension*

$$f_x := \Big[k\big(\mathfrak{m}_{X,x}\big) : \phi_\star k\big(\mathfrak{m}_{Y,\phi(x)}\big)\Big]$$

Observe that if x and x' are topologically indistinguishable (equivalently Galois conjugate, or equal in X_0) then $e_x = e_{x'}$ and $f_x = f_{x'}$.

Note when k is algebraically closed then $f_x = 1$ since $k(x), k(y)$ are finite and therefore trivial extensions of k.

Proposition 6.2.133 (Ramification Formula)
Let $\phi : X \to Y$ be a morphism of regular curves which is finite (e.g. X and Y are complete (6.2.130)). For $y \in Y$ then

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$$\deg(\phi) = \sum_{x_0 \in \phi_0^{-1}(\{y_0\})} e_x f_x$$

Proof. Let V be an affine neighbourhood of y, and define $U = \phi^{-1}(V)$. Then by assumption U is affine and $\phi^\sharp : \mathcal{O}_Y(V) \to \mathcal{O}_X(U)$ is finite (and integral (3.23.4)).

Define $L \coloneqq k(Y)$, $B \coloneqq \mathcal{O}_Y(V)$, $K \coloneqq \phi_\star(k(X))$ and $A \coloneqq \phi_\star(\mathcal{O}_X(U))$, where $\phi_\star : k(X) \hookrightarrow k(Y)$ is the canonical embedding

Then by (6.2.15) L is the quotient field of B, and K is the quotient field of A. By (6.2.101) (6.2.54) both B and A are Dedekind domains. B is finite and integral over A, and therefore B is the integral closure of A, being integrally closed. Therefore we are in the situation of (3.31.12).

By (6.2.21) there is a bijection between the maximal ideals of B (resp. A) and the points of U_0 (resp. V_0). And by (...) $\phi_0(x_0) = y_0 \iff \mathfrak{M}_{U,x} \cap A = \mathfrak{M}_{V,y}$.

By (3.31.19) the notion of ramification index and residue degree coincide with the same notions for Dedekind domains. Therefore the result follows from the corresponding result for finite morphisms of Dedekind domains (3.31.20). ∎

Proposition 6.2.134 (Degree of Principal Divisor is Zero)
Let X be a complete regular curve and $f \in k(X)$ a non-constant rational function, then

$$\deg((f)) = 0$$

i.e. the degree of a principal divisor is zero.

Proof. By (6.2.129) f corresponds to a non-constant and therefore surjective regular morphism $\phi_f : X \to \mathbb{P}^1_k \xrightarrow{\sim} \mathbb{A}^1_k \cup \{\infty\}$. From the construction we see that $v_x(f) > 0$ if and only if $\phi_f(x) = 0$ and $v_x(f) < 0$ if and only if $\phi_f(x) = \infty$. Therefore by definition

$$(f) = \sum_{\substack{x_0 \in X_0 \\ \phi(x_0)=0}} v_{x_0}(f) x_0 + \sum_{\substack{x_0 \in X_0 \\ \phi(x_0)=\infty}} v_{x_0}(f) x_0$$

We have canonical isomorphism $\mathcal{O}_{\mathbb{P}^1_k,0} \xrightarrow{\sim} k[X]_{(X)}$ and when $\phi_f(x) = 0$ the corresponding morphisms of stalks is given by $X \to f_x$, therefore the ramification index is $e_x = v_x(f)$. Similarly when $\phi_f(x) = \infty$ we have $e_x = v_x(f^{-1}) = -v_x(f)$. It's evident that $k\left(\mathfrak{m}_{\mathbb{P}^1_k,0}\right) = k$, whence $f_{x_0} = [k(x_0) : k]$. Therefore by definition of degree

$$\deg((f)) = \sum_{\substack{x_0 \in X_0 \\ \phi(x_0)=0}} e_{x_0} f_{x_0} - \sum_{\substack{x_0 \in X_0 \\ \phi(x_0)=\infty}} e_{x_0} f_{x_0}$$

As X is complete, and so is $\mathbb{P}^1_k$ (6.2.95) then ϕ_f is finite by (6.2.130) and we are done by (6.2.133) since both sums are equal to $\deg(\phi_f)$. ∎

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